I. Training Environment, Hyperparameters, and RTT Settings

A. Training Environment

All experiments for this work are done on a single NVIDIA A800 GPU on Ubuntu 20.04, python 3.6, torch 1.7, and transformers 4.2.1.

B. Hyperparameters and RTT Settings

We report the important hyperparameters and datasets of the pre-trained encoder, JTCSE-BERT/RoBERTa(D, UC D), in Table I-1.

JTCSE is divided into two stages of training: the first stage is the pre-training encoder, and the second stage is the training of the twin-tower model, which adds tensor norm constraints. For the first stage, we set the learning rate search space from 1e-5 to 1.5e-5, and the step size is set to 1e-6 or 1.5e-6 based on BERT or RoBERTa. The second stage must reduce the learning rate, which is uniformly set to 1.5e-5. JTCSE-D and JTCSE-UC D are cumulative training, and the learning rate search space and step size are the same as the first stage of JTCSE. The batch size is set to 64, 128, 256, and we do not tune for random seeds, which is uniformly set to 42. We evaluate the performance of the model on the STS-B validation set once every one hundred steps of training, as a checkpoint saved metric.

For the training set, we uniformly employ Wiki1M and unlabelled SICKR. For the selection of RTT languages, four randomly selected languages, en-fr-en, en-ru-en, en-nl-en, and en-zh-en, are used for the first stage of the training of the encoder for JTCSE-BERT/RoBERTa. For the other models in the JTCSE series, we uniformly do not use RTT Data Augmentation.

II. Details of Text Retrieval Task Results

In this section, we report the details of the 45 text retrieval zero-shot experiments for each metric, and we report the results for MAP@1/5/10, MRR@1/5/10, NDCG@1/5/10, PRECISION@1/5/10, and RECALL@1/5/10 in Tables II-1 through II-15, respectively.

TABLE I-1
We report important hyperparameters for model training.

Model	Learning Rate	Batch Size	Datasets
TNCSE-BERT			
Encoder I	1.5e-5	64	Wiki1M+Unlabelled en-fr-en RTT
Encoder II	2e-5	64	Wiki1M+Unlabelled en-ru-en RTT
JTCSE-BERT	1.3e-6	64	Wiki1M+Unlabelled
+D	1.5e-5	64	Wiki1M+Unlabelled
+UC D	1.5e-5	64	Wiki1M+Unlabelled
TNCSE-RoBERTa			
Encoder I	1.5e-5	64	Wiki1M+Unlabelled en-nl-en RTT
Encoder II	1.25e-5	64	Wiki1M+Unlabelled en-zh-en RTT
JTCSE-RoBERTa	1.3e-6	64	Wiki1M+Unlabelled
+D	1e-5	64	Wiki1M+Unlabelled
+UC D	1e-5	256	Wiki1M+Unlabelled

TABLE II-1
THIS TABLE REPORTS THE PERFORMANCE OF JTCSE AND DERIVED MODELS AND OTHER BASELINES THROUGH THE MTEB PACKAGE ON 45 TEXT RETRIEVAL TASKS EVALUATED WITH THE METRIC **MAP@1**.

Tasks	Sim∼	ESim∼	Diff∼	Info∼	SN∼	Whinen∼	Rank∼	TN∼	JT∼	JT∼ D	JT∼ UCD
AILACasedocs	7.91	**9.23**	5.53	4.49	4.51	3.37	5.46	5.96	7.70	6.03	6.16
AlloprofRetrieval	2.38	3.37	1.34	1.17	0.43	0.60	1.99	2.33	3.45	**3.67**	3.02
CQADupstackAndroidRetrieval	12.78	13.98	14.83	18.32	1.27	17.16	14.30	13.33	15.87	15.99	**19.14**
ClimateFEVER	4.58	2.50	4.22	2.71	1.10	4.57	**4.61**	3.54	3.43	3.85	4.32
ClimateFEVERHardNegatives	**4.93**	2.54	4.32	2.80	1.13	4.57	4.85	3.67	3.46	4.04	4.34
CodeFeedbackMT	19.30	23.42	22.87	22.56	12.77	21.91	19.99	20.25	21.77	21.35	**23.59**
CodeTransOceanDL	**8.33**	7.78	**8.33**	**8.33**	**8.33**	**8.33**	7.78	**8.33**	**8.33**	**8.33**	**8.33**
CosQA	1.60	1.80	2.20	**3.80**	0.00	1.60	1.80	1.20	2.60	2.80	2.20
CrossLingualSemanticDiscriminationWMT19	10.32	13.92	5.64	6.45	6.11	6.52	4.35	**16.63**	14.60	13.37	8.83
CrossLingualSemanticDiscriminationWMT21	15.45	21.17	12.99	10.08	9.63	11.31	11.09	**25.42**	22.06	19.49	16.35
DBPedia-PLHardNegatives	**1.96**	1.08	0.80	0.70	0.05	1.20	0.93	0.79	1.11	1.11	0.80
DanFeverRetrieval	19.32	18.41	19.59	19.63	10.70	**20.84**	19.39	14.19	18.83	18.59	19.16
DuRetrieval	0.22	0.70	0.46	0.73	0.03	0.67	0.57	0.71	**1.03**	0.96	1.00
EcomRetrieval	1.10	0.90	1.40	**2.50**	0.20	**2.50**	1.70	1.30	1.70	2.10	2.30
FEVER	14.35	4.80	12.13	8.43	2.21	**15.30**	11.73	9.48	8.03	9.26	10.43
GeorgianFAQRetrieval	1.79	1.64	1.75	2.11	0.78	2.18	1.44	1.21	2.22	**2.61**	2.30
GerDaLIR	0.49	0.56	0.43	0.34	0.22	0.39	0.36	0.37	0.61	0.67	**0.67**
GerDaLIRSmall	1.22	1.38	0.92	0.85	0.51	0.96	0.87	0.95	1.51	**1.61**	1.58
GermanQuAD-Retrieval	24.09	**29.90**	18.24	14.75	5.31	11.16	22.28	20.83	28.86	28.00	27.27
GreekCivicsQA	2.95	**3.93**	2.95	0.98	0.49	2.70	2.46	2.21	3.19	**3.93**	**3.93**
LEMBNarrativeQARetrieval	7.13	6.27	7.55	6.24	1.35	7.73	7.01	8.13	**8.14**	7.63	7.94
LEMBSummScreenFDRetrieval	32.44	36.61	**39.29**	32.44	11.31	38.99	27.98	37.80	36.31	37.80	38.39
LeCaRDv2	0.92	0.72	1.01	**1.20**	0.92	0.77	0.84	0.59	0.84	0.78	0.69
LegalBenchCorporateLobbying	55.59	64.41	62.65	62.65	25.29	65.88	59.41	65.59	**67.06**	63.53	**67.06**
MSMARCO-PL	0.02	0.04	0.05	0.16	0.00	0.02	0.01	0.10	0.13	**0.21**	0.11
MedicalQARetrieval	10.50	9.91	11.38	14.01	0.49	**17.97**	12.26	11.82	13.09	13.18	13.48
MedicalRetrieval	0.20	0.30	0.40	**1.10**	0.00	0.40	0.50	0.30	0.30	0.40	0.60
NFCorpus	0.73	1.18	1.27	1.26	0.06	1.26	0.91	1.59	**1.76**	1.44	1.36
NFCorpus-PL	0.38	1.03	0.67	0.23	0.10	0.48	0.63	0.48	**1.12**	0.80	0.48
NQ-PL	0.24	0.20	0.15	**0.38**	0.03	0.27	0.14	0.14	0.26	0.33	0.20
NanoNFCorpusRetrieval	0.38	1.64	2.59	1.59	0.00	**3.74**	0.49	3.59	1.63	1.60	1.88
NanoNQRetrieval	14.00	4.00	**16.00**	11.00	0.00	**16.00**	**16.00**	10.00	10.00	15.00	12.00
NanoTouche2020Retrieval	**2.97**	2.55	2.33	1.31	0.49	2.20	2.45	2.29	2.38	2.33	2.27
NeuCLIR2023RetrievalHardNegatives	0.00	**0.30**	0.01	0.00	0.00	0.00	0.00	0.02	0.01	0.01	0.00
PublicHealthQA	8.14	**12.79**	**12.79**	3.49	3.49	10.47	8.14	9.30	10.47	9.30	9.30
SCIDOCS	1.26	1.58	1.76	1.60	0.37	1.62	1.49	1.42	1.73	1.89	**2.09**
SciFact-PL	1.67	**9.03**	4.28	2.33	0.67	3.17	1.33	6.67	5.53	5.94	4.78
SlovakSumRetrieval	8.50	**24.33**	10.83	5.67	3.33	8.17	8.33	12.17	23.83	23.67	21.50
TRECCOVID	0.10	0.10	0.10	**0.13**	0.03	0.11	0.09	0.08	0.12	0.11	0.11
TV2Nordretrieval	8.30	**23.19**	10.50	6.74	3.03	7.72	7.28	13.72	20.51	21.44	18.21
TurHistQuadRetrieval	3.32	4.15	3.66	3.47	1.17	3.71	2.78	3.08	5.27	**5.57**	5.03
TwitterHjerneRetrieval	3.93	4.57	3.33	3.61	3.36	4.00	3.14	**4.74**	4.42	4.32	4.36
VideoRetrieval	4.40	4.10	5.40	**6.20**	2.20	5.50	4.90	3.50	4.40	4.30	5.20
VieQuADRetrieval	0.93	2.39	1.64	0.76	0.85	1.15	1.27	1.51	2.10	**2.42**	2.10
XMarket	1.06	0.80	1.32	**2.14**	0.17	1.64	0.64	0.70	1.21	1.21	1.41
45 Avg. MAP@1	7.16	8.43	7.60	6.70	2.77	7.57	6.80	7.82	**8.73**	**8.73**	8.58

TABLE II-2

THIS TABLE REPORTS THE PERFORMANCE OF JTCSE AND DERIVED MODELS AND OTHER BASELINES THROUGH THE MTEB PACKAGE ON 45 TEXT RETRIEVAL TASKS EVALUATED WITH THE METRIC **MAP@5**.

Tasks	Sim∼	ESim∼	Diff∼	Info∼	SN∼	Whinen∼	Rank∼	TN∼	JT∼	JT∼ D	JT∼ UCD
AILACasedocs	**12.88**	12.60	8.17	11.99	8.86	7.17	9.51	10.47	12.52	12.03	12.86
AlloprofRetrieval	3.77	**5.78**	2.17	2.16	0.74	1.13	3.17	3.59	5.41	5.45	4.85
CQADupstackAndroidRetrieval	17.56	17.96	19.14	23.66	1.93	21.93	18.26	17.50	20.05	20.37	**24.61**
ClimateFEVER	7.36	4.60	6.90	4.26	1.70	7.42	**7.79**	6.61	5.76	6.56	7.47
ClimateFEVERHardNegatives	7.96	5.03	7.23	4.58	1.68	7.68	**8.12**	6.96	6.18	6.86	7.71
CodeFeedbackMT	22.55	26.99	26.32	26.00	15.42	25.51	23.47	23.74	25.41	25.06	**27.54**
CodeTransOceanDL	14.28	14.61	15.39	13.57	16.31	**16.88**	12.91	15.88	15.49	14.57	15.21
CosQA	2.77	2.95	3.60	**6.34**	0.12	2.55	3.34	1.79	3.63	3.88	4.02
CrossLingualSemanticDiscriminationWMT19	14.92	18.61	8.63	9.21	8.96	9.74	6.83	**22.90**	19.68	18.40	12.79
CrossLingualSemanticDiscriminationWMT21	21.82	29.52	17.99	16.03	13.39	17.02	16.03	**35.21**	30.95	28.56	23.23
DBPedia-PLHardNegatives	**2.77**	1.87	2.04	1.60	0.20	2.07	2.12	1.43	2.18	2.32	2.05
DanFeverRetrieval	22.23	21.95	22.91	22.05	12.06	**23.74**	22.50	16.83	22.35	22.23	22.83
DuRetrieval	0.51	1.16	0.87	1.27	0.07	1.18	0.90	1.12	1.60	1.63	**1.64**
EcomRetrieval	1.79	1.49	2.32	**3.63**	0.55	3.32	2.68	2.12	2.47	2.84	3.33
FEVER	19.84	7.08	17.54	11.16	3.28	**21.24**	16.50	13.53	11.61	13.21	14.78
GeorgianFAQRetrieval	2.55	2.28	2.75	3.06	1.20	3.10	2.15	2.03	3.30	**3.68**	3.58
GerDaLIR	0.64	0.78	0.54	0.48	0.27	0.49	0.47	0.54	0.84	0.88	**0.90**
GerDaLIRSmall	1.68	1.85	1.22	1.15	0.72	1.24	1.22	1.38	2.01	**2.07**	2.06
GermanQuAD-Retrieval	30.38	**37.34**	23.93	19.69	8.03	15.68	29.14	27.35	36.32	35.50	34.56
GreekCivicsQA	5.34	5.52	4.84	1.88	1.00	5.43	4.16	3.23	6.05	**6.83**	6.77
LEMBNarrativeQARetrieval	9.31	8.32	9.91	8.48	2.20	10.09	9.38	**10.86**	10.71	10.17	10.41
LEMBSummScreenFDRetrieval	45.21	47.31	50.31	43.69	18.00	**51.43**	38.88	49.22	48.48	48.54	48.82
LeCaRDv2	2.43	1.68	2.42	**2.85**	2.58	1.73	2.39	1.34	2.09	1.72	1.69
LegalBenchCorporateLobbying	66.16	74.68	72.14	71.88	34.07	74.01	68.72	74.88	75.56	72.94	**76.32**
MSMARCO-PL	0.03	0.10	0.11	**0.29**	0.00	0.04	0.03	0.12	0.17	0.23	0.17
MedicalQARetrieval	16.34	14.73	17.13	20.76	1.10	**25.91**	18.23	18.20	20.08	20.00	21.01
MedicalRetrieval	0.30	0.37	0.57	**1.31**	0.10	0.57	0.56	0.40	0.43	0.55	0.75
NFCorpus	1.81	1.95	2.48	2.49	0.18	**3.34**	1.71	3.09	3.16	2.91	3.03
NFCorpus-PL	0.88	1.61	1.24	0.50	0.17	1.34	0.90	0.76	**1.98**	1.80	1.46
NQ-PL	0.46	0.47	0.32	0.53	0.03	0.35	0.27	0.17	0.48	**0.57**	0.39
NanoNFCorpusRetrieval	0.75	2.51	3.85	2.46	0.07	4.90	1.24	**5.01**	4.01	4.01	4.16
NanoNQRetrieval	22.87	13.13	25.00	18.97	1.00	23.73	**25.03**	17.40	21.37	24.73	23.37
NanoTouche2020Retrieval	7.43	6.31	7.82	4.09	0.72	8.76	7.46	6.78	8.17	**9.62**	7.54
NeuCLIR2023RetrievalHardNegatives	0.01	**0.33**	0.01	0.00	0.00	0.00	0.01	0.18	0.02	0.08	0.05
PublicHealthQA	14.01	**18.84**	18.10	7.17	7.15	15.95	13.02	15.58	17.23	16.63	18.24
SCIDOCS	2.53	3.16	3.32	3.15	0.51	3.37	2.67	2.71	3.34	3.54	**3.82**
SciFact-PL	2.97	**10.93**	5.07	3.08	1.01	3.72	2.34	8.68	8.64	8.58	7.75
SlovakSumRetrieval	11.95	28.71	15.47	7.81	5.16	11.53	11.38	15.95	**28.98**	28.63	26.42
TRECCOVID	0.30	0.27	0.32	**0.42**	0.07	0.27	0.30	0.29	0.30	0.31	0.34
TV2Nordretrieval	10.72	**28.51**	13.28	8.97	4.14	9.76	9.77	17.91	25.53	26.59	23.46
TurHistQuadRetrieval	5.05	6.70	5.43	5.21	1.87	5.72	4.21	4.28	7.90	**8.24**	7.07
TwitterHjerneRetrieval	7.49	7.60	6.39	6.96	5.61	6.43	6.80	**7.75**	6.73	6.18	6.80
VideoRetrieval	5.64	5.48	6.86	**8.07**	3.08	7.11	6.50	4.90	5.83	5.95	6.84
VieQuADRetrieval	1.79	**4.48**	2.86	1.35	1.38	1.87	2.14	2.34	3.44	4.02	3.35
XMarket	1.85	1.45	2.28	**3.55**	0.33	2.68	1.44	1.17	2.10	2.11	2.37
45 Avg. MAP@5	10.04	11.32	10.38	9.28	4.16	10.42	9.48	10.76	12.01	**12.03**	11.96

TABLE II-3
THIS TABLE REPORTS THE PERFORMANCE OF JTCSE AND DERIVED MODELS AND OTHER BASELINES THROUGH THE MTEB PACKAGE ON 45 TEXT RETRIEVAL TASKS EVALUATED WITH THE METRIC **MAP@10**.

Tasks	Sim∼	ESim∼	Diff∼	Info∼	SN∼	Whinen∼	Rank∼	TN∼	JT∼	JT∼ D	JT∼ UCD
AILACasedocs	14.90	**15.62**	10.16	13.67	10.33	8.34	12.04	12.01	13.73	13.09	14.54
AlloprofRetrieval	4.12	**6.25**	2.45	2.49	0.84	1.32	3.57	3.98	5.92	6.00	5.37
CQADupstackAndroidRetrieval	18.45	18.76	19.95	24.79	2.08	23.15	19.06	18.46	21.03	21.27	**25.56**
ClimateFEVER	8.18	5.22	7.66	4.71	1.92	8.24	**8.62**	7.45	6.65	7.48	8.41
ClimateFEVERHardNegatives	8.81	5.78	8.12	5.16	1.93	8.63	**9.04**	7.92	7.10	7.93	8.62
CodeFeedbackMT	23.07	27.53	26.81	26.52	15.87	25.97	23.94	24.25	25.94	25.60	**28.10**
CodeTransOceanDL	18.95	19.37	20.38	17.58	21.06	**21.66**	16.72	21.37	19.98	18.63	19.44
CosQA	3.15	3.42	4.02	**7.05**	0.17	2.91	3.61	2.03	4.07	4.47	4.51
CrossLingualSemanticDiscriminationWMT19	15.55	19.30	9.29	9.78	9.36	10.24	7.37	**23.80**	20.39	19.00	13.44
CrossLingualSemanticDiscriminationWMT21	22.69	30.52	18.76	16.64	14.53	17.88	17.06	**36.30**	31.67	29.55	24.20
DBPedia-PLHardNegatives	**3.31**	2.44	2.53	1.95	0.22	2.43	2.55	1.80	2.73	2.83	2.55
DanFeverRetrieval	22.61	22.38	23.27	22.34	12.26	**24.11**	22.82	17.13	22.80	22.68	23.28
DuRetrieval	0.60	1.30	0.93	1.40	0.07	1.28	0.99	1.22	1.72	1.77	**1.80**
EcomRetrieval	1.96	1.66	2.51	**3.92**	0.57	3.49	2.84	2.34	2.63	2.98	3.45
FEVER	20.79	7.72	18.49	11.70	3.53	**22.29**	17.33	14.34	12.29	13.98	15.67
GeorgianFAQRetrieval	2.71	2.43	2.94	3.27	1.30	3.30	2.35	2.18	3.56	**3.90**	3.82
GerDaLIR	0.68	0.82	0.55	0.50	0.29	0.52	0.52	0.58	0.88	0.93	**0.94**
GerDaLIRSmall	1.80	1.96	1.28	1.20	0.77	1.31	1.32	1.47	2.13	2.17	**2.18**
GermanQuAD-Retrieval	31.42	**38.25**	24.74	20.66	8.57	16.69	29.97	28.33	37.36	36.39	35.58
GreekCivicsQA	6.24	6.22	5.55	2.19	1.22	6.03	4.75	3.73	6.72	**7.67**	7.64
LEMBNarrativeQARetrieval	9.82	8.82	10.41	8.97	2.53	10.55	9.93	**11.52**	11.30	10.77	10.98
LEMBSummScreenFDRetrieval	46.75	48.67	51.77	45.63	19.35	**53.11**	40.40	51.02	50.33	50.16	50.29
LeCaRDv2	3.35	2.13	3.29	**3.96**	3.48	2.36	3.14	1.70	2.85	2.12	2.24
LegalBenchCorporateLobbying	66.71	75.45	72.96	72.93	34.84	74.76	69.61	75.44	76.12	73.42	**76.90**
MSMARCO-PL	0.09	0.14	0.15	**0.39**	0.00	0.05	0.06	0.13	0.19	0.28	0.20
MedicalQARetrieval	17.46	15.71	18.35	22.07	1.27	**27.01**	19.20	19.25	21.43	21.36	22.29
MedicalRetrieval	0.35	0.39	0.63	**1.35**	0.11	0.59	0.64	0.48	0.52	0.59	0.82
NFCorpus	2.19	2.45	2.90	2.93	0.24	**3.83**	2.10	3.44	3.78	3.51	3.54
NFCorpus-PL	1.07	1.86	1.55	0.58	0.19	1.52	0.99	0.98	**2.22**	1.99	1.67
NQ-PL	0.50	0.54	0.37	0.56	0.03	0.39	0.29	0.21	0.54	**0.62**	0.44
NanoNFCorpusRetrieval	0.88	2.90	4.13	2.76	0.43	5.35	1.62	**5.63**	4.37	4.30	4.50
NanoNQRetrieval	25.55	14.14	26.28	20.88	1.00	25.56	**27.21**	19.16	22.57	27.21	25.44
NanoTouche2020Retrieval	11.14	9.86	10.72	5.70	1.00	13.06	11.44	9.63	11.58	**13.21**	12.12
NeuCLIR2023RetrievalHardNegatives	0.05	**0.51**	0.02	0.00	0.00	0.02	0.02	0.22	0.19	0.22	0.10
PublicHealthQA	15.30	**20.09**	19.37	8.15	8.05	17.53	13.65	16.41	18.20	17.37	18.86
SCIDOCS	2.88	3.69	3.86	3.61	0.55	3.79	3.01	3.25	3.97	4.06	**4.39**
SciFact-PL	3.38	**11.64**	5.35	3.20	1.01	3.83	2.68	8.78	9.06	8.86	8.33
SlovakSumRetrieval	12.80	29.59	16.52	8.47	5.57	12.52	12.13	16.87	**29.86**	29.51	27.50
TRECCOVID	0.45	0.43	0.50	**0.69**	0.07	0.40	0.48	0.44	0.46	0.48	0.52
TV2Nordretrieval	11.27	**29.44**	13.84	9.41	4.42	10.21	10.26	18.58	26.48	27.48	24.42
TurHistQuadRetrieval	5.45	7.30	5.93	5.56	2.06	6.17	4.65	4.52	8.48	**8.83**	7.57
TwitterHjerneRetrieval	7.92	8.14	7.25	7.64	6.09	7.29	7.52	**8.86**	7.46	6.69	7.35
VideoRetrieval	5.97	5.76	7.12	**8.56**	3.26	7.44	6.90	5.20	6.20	6.28	7.11
VieQuADRetrieval	2.12	**5.08**	3.28	1.50	1.55	2.08	2.34	2.57	3.93	4.51	3.72
XMarket	2.12	1.67	2.55	**3.97**	0.38	3.02	1.69	1.31	2.37	2.37	2.67
45 Avg. MAP@10	10.79	12.07	11.10	9.93	4.54	11.16	10.19	11.47	12.75	**12.77**	12.73

TABLE II-4

THIS TABLE REPORTS THE PERFORMANCE OF JTCSE AND DERIVED MODELS AND OTHER BASELINES THROUGH THE MTEB PACKAGE ON 45 TEXT RETRIEVAL TASKS EVALUATED WITH THE METRIC **MRR@1**.

Tasks	Sim∼	ESim∼	Diff∼	Info∼	SN∼	Whinen∼	Rank∼	TN∼	JT∼	JT∼ D	JT∼ UCD
AILACasedocs	**24.00**	**24.00**	12.00	16.00	14.00	10.00	14.00	16.00	16.00	16.00	18.00
AlloprofRetrieval	2.37	3.37	1.34	1.17	0.43	0.60	1.99	2.33	3.45	**3.67**	3.02
CQADupstackAndroidRetrieval	16.88	17.74	18.17	23.18	1.86	21.75	18.45	17.74	20.46	20.46	**24.61**
ClimateFEVER	9.71	5.80	8.73	6.19	2.54	9.84	9.45	8.21	7.95	8.86	**9.90**
ClimateFEVERHardNegatives	**10.00**	6.00	8.60	5.80	2.70	9.60	9.80	8.40	7.80	9.20	9.80
CodeFeedbackMT	19.30	23.42	22.87	22.56	12.77	21.91	19.99	20.25	21.77	21.35	**23.59**
CodeTransOceanDL	**8.33**	7.78	**8.33**	7.22	**8.33**	**8.33**	7.22	**8.33**	**8.33**	**8.33**	**8.33**
CosQA	1.20	1.80	1.80	**2.60**	0.20	1.20	0.80	0.60	2.20	2.00	2.00
CrossLingualSemanticDiscriminationWMT19	10.32	13.99	5.63	6.45	6.11	6.52	4.34	**16.63**	14.60	13.37	8.83
CrossLingualSemanticDiscriminationWMT21	15.45	21.16	12.99	10.08	9.63	11.31	11.09	**25.42**	22.06	19.48	16.35
DBPedia-PLHardNegatives	14.75	13.75	10.75	11.50	2.25	9.50	10.25	12.00	14.25	**15.50**	13.50
DanFeverRetrieval	19.32	18.41	19.60	19.63	10.70	**20.84**	19.39	14.18	18.83	18.59	19.16
DuRetrieval	1.10	2.30	1.70	2.60	0.20	2.25	1.95	2.15	3.10	2.95	**3.20**
EcomRetrieval	1.10	0.90	1.40	**2.50**	0.20	**2.50**	1.70	1.30	1.70	2.10	2.30
FEVER	15.18	5.10	12.84	8.94	2.30	**16.19**	12.41	10.22	8.66	10.02	11.21
GeorgianFAQRetrieval	1.79	1.64	1.75	2.11	0.78	2.18	1.44	1.21	2.22	**2.61**	2.30
GerDaLIR	0.54	0.63	0.46	0.39	0.25	0.42	0.40	0.41	0.67	**0.74**	**0.74**
GerDaLIRSmall	1.37	1.53	1.02	0.95	0.58	1.05	0.95	1.06	1.64	**1.76**	1.75
GermanQuAD-Retrieval	24.09	**29.90**	18.24	14.75	5.31	11.16	22.28	20.83	28.86	27.99	27.27
GreekCivicsQA	2.95	**3.93**	2.95	0.98	0.49	2.70	2.46	2.21	3.19	**3.93**	**3.93**
LEMBNarrativeQARetrieval	7.13	6.27	7.55	6.24	1.35	7.73	7.01	8.13	**8.13**	7.63	7.94
LEMBSummScreenFDRetrieval	32.44	36.61	**39.29**	32.44	11.31	38.99	27.98	37.80	36.31	37.80	38.39
LeCaRDv2	24.53	18.24	25.16	**27.67**	25.16	20.75	22.64	16.35	22.01	20.13	18.87
LegalBenchCorporateLobbying	55.59	64.41	62.65	62.65	25.29	65.88	59.41	65.59	**67.06**	63.53	**67.06**
MSMARCO-PL	2.33	4.65	**6.98**	**6.98**	0.00	2.33	2.33	2.33	4.65	**6.98**	4.65
MedicalQARetrieval	10.45	9.96	11.43	14.01	0.49	**17.97**	12.26	11.82	13.18	13.23	13.53
MedicalRetrieval	0.20	0.30	0.40	**1.10**	0.00	0.40	0.50	0.30	0.30	0.40	0.60
NFCorpus	15.48	18.27	17.03	19.20	2.79	18.27	13.93	19.50	**20.74**	**20.74**	**20.74**
NFCorpus-PL	6.19	**8.67**	5.88	2.79	2.48	5.26	5.57	4.95	**8.67**	7.74	5.88
NQ-PL	0.32	0.26	0.17	**0.41**	0.03	0.35	0.17	0.17	0.32	0.38	0.23
NanoNFCorpusRetrieval	14.00	24.00	22.00	20.00	2.00	**28.00**	16.00	20.00	26.00	24.00	**28.00**
NanoNQRetrieval	14.00	4.00	**16.00**	12.00	0.00	**16.00**	**16.00**	10.00	10.00	**16.00**	12.00
NanoTouche2020Retrieval	**42.86**	34.69	30.61	20.41	8.16	28.57	28.57	30.61	30.61	30.61	32.65
NeuCLIR2023RetrievalHardNegatives	0.00	**5.41**	1.35	0.00	0.00	0.00	0.00	2.70	1.35	1.35	0.00
PublicHealthQA	8.14	**12.79**	**12.79**	3.49	3.49	10.47	8.14	9.30	10.47	9.30	9.30
SCIDOCS	6.20	7.70	8.60	7.80	1.80	7.90	7.30	6.90	8.50	9.30	**10.20**
SciFact-PL	1.67	**9.67**	4.67	2.33	0.67	3.33	1.33	7.00	6.33	6.33	5.33
SlovakSumRetrieval	8.50	**24.33**	10.83	5.67	3.33	8.17	8.33	12.17	23.83	23.67	21.50
TRECCOVID	42.00	36.00	40.00	**46.00**	10.00	40.00	34.00	36.00	**46.00**	44.00	40.00
TV2Nordretrieval	8.30	**23.19**	10.50	6.74	3.03	7.71	7.28	13.72	20.51	21.44	18.21
TurHistQuadRetrieval	6.64	8.30	7.32	7.03	2.34	7.52	5.66	6.15	10.55	**11.13**	10.06
TwitterHjerneRetrieval	14.10	**16.67**	12.82	14.10	11.54	12.82	11.54	15.38	14.10	14.10	15.38
VideoRetrieval	4.40	4.10	5.40	**6.30**	2.30	5.50	4.90	3.60	4.40	4.30	5.20
VieQuADRetrieval	1.86	4.79	3.27	1.51	1.71	2.29	2.54	3.03	4.20	**4.83**	4.20
XMarket	4.41	4.06	5.55	**8.82**	1.49	6.94	3.67	3.59	5.47	5.20	6.07
45 Avg. MRR@1	11.81	13.12	11.99	11.14	4.50	11.84	10.61	11.92	13.59	**13.62**	13.46

TABLE II-5

THIS TABLE REPORTS THE PERFORMANCE OF JTCSE AND DERIVED MODELS AND OTHER BASELINES THROUGH THE MTEB PACKAGE ON 45 TEXT RETRIEVAL TASKS EVALUATED WITH THE METRIC **MRR@5**.

Tasks	Sim∼	ESim∼	Diff∼	Info∼	SN∼	Whinen∼	Rank∼	TN∼	JT∼	JT∼ D	JT∼ UCD
AILACasedocs	**31.70**	28.83	18.27	28.60	23.07	18.63	21.70	23.40	26.30	27.93	30.20
AlloprofRetrieval	3.77	**5.78**	2.17	2.16	0.74	1.13	3.17	3.59	5.41	5.44	4.85
CQADupstackAndroidRetrieval	22.07	22.08	23.23	29.17	2.94	26.60	22.49	22.31	24.87	25.25	**30.16**
ClimateFEVER	15.38	10.27	14.32	9.18	4.04	15.76	**16.38**	14.28	12.75	14.49	16.34
ClimateFEVERHardNegatives	16.23	11.32	14.70	9.53	4.19	16.27	**17.01**	14.88	13.45	15.09	16.56
CodeFeedbackMT	22.55	26.99	26.32	26.00	15.42	25.51	23.47	23.74	25.41	25.06	**27.54**
CodeTransOceanDL	14.80	15.40	16.37	13.60	16.89	**17.50**	12.34	16.92	16.31	15.81	16.18
CosQA	2.16	2.50	2.95	**5.25**	0.20	2.25	2.12	1.12	3.37	3.26	3.58
CrossLingualSemanticDiscriminationWMT19	14.92	18.64	8.64	9.21	8.96	9.74	6.83	**22.90**	19.68	18.40	12.79
CrossLingualSemanticDiscriminationWMT21	21.82	29.52	17.99	16.03	13.39	17.02	16.03	**35.21**	30.95	28.56	23.23
DBPedia-PLHardNegatives	20.19	18.63	17.01	17.65	3.35	16.02	16.83	15.57	20.00	**21.38**	20.30
DanFeverRetrieval	22.23	21.95	22.92	22.05	12.06	**23.74**	22.50	16.83	22.35	22.23	22.83
DuRetrieval	1.72	3.05	2.47	3.79	0.35	3.26	2.63	3.12	4.23	4.33	**4.36**
EcomRetrieval	1.79	1.49	2.32	**3.63**	0.55	3.32	2.68	2.12	2.47	2.84	3.33
FEVER	21.06	7.50	18.50	11.84	3.46	**22.43**	17.44	14.48	12.42	14.16	15.78
GeorgianFAQRetrieval	2.55	2.28	2.75	3.06	1.20	3.10	2.14	2.03	3.30	**3.68**	3.58
GerDaLIR	0.71	0.86	0.58	0.54	0.31	0.54	0.52	0.60	0.92	0.97	**0.99**
GerDaLIRSmall	1.88	2.02	1.34	1.28	0.81	1.36	1.34	1.53	2.21	2.26	**2.28**
GermanQuAD-Retrieval	30.38	**37.34**	23.93	19.69	8.03	15.68	29.14	27.35	36.32	35.50	34.56
GreekCivicsQA	5.34	5.52	4.84	1.88	1.00	5.43	4.16	3.23	6.05	**6.83**	6.77
LEMBNarrativeQARetrieval	9.31	8.32	9.91	8.48	2.20	10.09	9.38	**10.86**	10.71	10.17	10.41
LEMBSummScreenFDRetrieval	45.21	47.31	50.31	43.69	18.00	**51.43**	38.88	49.22	48.48	48.54	48.82
LeCaRDv2	30.22	22.67	30.56	**32.98**	30.44	25.77	27.88	19.25	27.90	23.44	22.91
LegalBenchCorporateLobbying	66.16	74.68	72.14	71.88	34.07	74.00	68.72	74.88	75.56	72.94	**76.32**
MSMARCO-PL	2.91	6.47	8.14	**11.43**	0.00	3.10	2.91	4.46	6.67	9.38	6.98
MedicalQARetrieval	16.30	14.78	17.15	20.76	1.10	**25.90**	18.22	18.20	20.16	20.02	21.04
MedicalRetrieval	0.30	0.37	0.57	**1.31**	0.10	0.57	0.56	0.40	0.43	0.55	0.75
NFCorpus	20.90	23.44	23.63	25.90	4.70	26.65	19.30	25.94	27.77	27.39	**27.81**
NFCorpus-PL	9.27	**12.76**	8.53	5.56	3.68	9.07	8.17	7.49	11.98	11.37	9.39
NQ-PL	0.57	0.59	0.38	0.57	0.03	0.43	0.34	0.22	0.55	**0.65**	0.45
NanoNFCorpusRetrieval	16.90	29.37	28.23	25.23	3.90	**34.97**	20.13	31.23	32.80	32.93	34.57
NanoNQRetrieval	23.07	13.97	25.33	20.30	1.00	23.73	25.23	17.40	21.70	**25.73**	23.70
NanoTouche2020Retrieval	**59.18**	49.49	49.59	37.79	12.93	49.12	46.53	48.54	49.97	52.18	50.92
NeuCLIR2023RetrievalHardNegatives	1.06	**6.98**	1.35	0.00	0.00	0.34	0.27	4.44	2.07	4.05	1.58
PublicHealthQA	14.01	**18.84**	18.10	7.17	7.15	15.95	13.02	15.58	17.23	16.63	18.24
SCIDOCS	10.32	12.40	12.89	12.24	2.27	12.85	11.34	10.62	12.65	13.53	**14.82**
SciFact-PL	3.02	**11.65**	5.51	3.13	1.07	3.98	2.49	9.17	9.39	9.11	8.07
SlovakSumRetrieval	11.95	28.71	15.47	7.81	5.16	11.53	11.38	15.95	**28.98**	28.63	26.42
TRECCOVID	50.23	48.20	48.40	**59.73**	20.70	46.90	45.73	47.20	53.70	51.30	50.03
TV2Nordretrieval	10.72	**28.51**	13.28	8.97	4.14	9.76	9.77	17.91	25.53	26.59	23.46
TurHistQuadRetrieval	9.67	13.11	10.45	10.10	3.64	11.06	8.08	8.40	15.24	**15.91**	13.66
TwitterHjerneRetrieval	19.98	21.58	17.20	**21.67**	16.62	17.65	19.06	21.20	19.04	18.10	19.32
VideoRetrieval	5.63	5.45	6.83	**8.08**	3.13	7.14	6.52	4.93	5.81	5.95	6.84
VieQuADRetrieval	3.50	**8.47**	5.52	2.66	2.74	3.74	4.14	4.61	6.57	7.60	6.45
XMarket	7.21	6.43	8.37	**12.56**	2.15	9.86	6.38	5.12	8.11	7.99	8.87
45 Avg. MRR@5	16.02	17.48	16.21	15.43	6.71	16.24	14.79	16.41	18.39	**18.54**	18.40

TABLE II-6

THIS TABLE REPORTS THE PERFORMANCE OF JTCSE AND DERIVED MODELS AND OTHER BASELINES THROUGH THE MTEB PACKAGE ON 45 TEXT RETRIEVAL TASKS EVALUATED WITH THE METRIC **MRR@10**.

Tasks	Sim∼	ESim∼	Diff∼	Info∼	SN∼	Whinen∼	Rank∼	TN∼	JT∼	JT∼ D	JT∼ UCD
AILACasedocs	**32.97**	31.01	20.72	29.39	24.86	20.25	24.06	25.22	27.09	28.69	30.53
AlloprofRetrieval	4.12	**6.25**	2.45	2.49	0.84	1.32	3.57	3.98	5.92	6.00	5.37
CQADupstackAndroidRetrieval	22.96	23.07	24.01	30.18	3.15	27.92	23.30	23.26	25.89	26.25	**31.00**
ClimateFEVER	16.72	11.35	15.66	10.10	4.42	17.13	17.57	15.84	14.29	15.94	**17.82**
ClimateFEVERHardNegatives	17.60	12.60	16.14	10.66	4.67	17.83	**18.36**	16.59	15.02	16.81	18.05
CodeFeedbackMT	23.07	27.53	26.81	26.52	15.87	25.97	23.94	24.25	25.94	25.60	**28.10**
CodeTransOceanDL	19.61	20.00	20.85	17.40	21.44	**22.16**	17.06	21.99	20.57	19.12	19.85
CosQA	2.63	3.15	3.34	**6.09**	0.30	2.56	2.65	1.45	3.69	3.87	4.10
CrossLingualSemanticDiscriminationWMT19	15.55	19.33	9.30	9.78	9.36	10.24	7.37	**23.80**	20.39	19.00	13.44
CrossLingualSemanticDiscriminationWMT21	22.69	30.52	18.76	16.64	14.52	17.88	17.06	**36.30**	31.67	29.55	24.20
DBPedia-PLHardNegatives	21.17	19.98	18.10	18.71	3.63	17.09	18.05	16.62	21.27	**22.24**	21.30
DanFeverRetrieval	22.61	22.38	23.27	22.34	12.26	**24.11**	22.82	17.13	22.80	22.68	23.28
DuRetrieval	1.94	3.27	2.61	4.03	0.38	3.46	2.81	3.31	4.44	4.59	**4.65**
EcomRetrieval	1.96	1.66	2.51	**3.92**	0.57	3.49	2.84	2.34	2.63	2.98	3.45
FEVER	22.05	8.19	19.49	12.39	3.71	**23.52**	18.32	15.31	13.13	14.96	16.70
GeorgianFAQRetrieval	2.71	2.43	2.93	3.27	1.30	3.30	2.35	2.18	3.56	**3.90**	3.82
GerDaLIR	0.76	0.90	0.60	0.56	0.33	0.56	0.57	0.64	0.97	1.03	**1.04**
GerDaLIRSmall	2.01	2.14	1.41	1.35	0.87	1.44	1.45	1.63	2.34	2.38	**2.42**
GermanQuAD-Retrieval	31.42	**38.25**	24.74	20.66	8.57	16.69	29.97	28.33	37.36	36.39	35.58
GreekCivicsQA	6.24	6.22	5.55	2.19	1.21	6.03	4.75	3.73	6.72	**7.67**	7.63
LEMBNarrativeQARetrieval	9.82	8.82	10.41	8.97	2.53	10.55	9.93	**11.52**	11.30	10.77	10.98
LEMBSummScreenFDRetrieval	46.75	48.67	51.77	45.63	19.35	**53.11**	40.40	51.02	50.33	50.15	50.29
LeCaRDv2	31.90	23.73	31.92	**34.42**	32.04	27.01	29.19	20.71	28.95	24.41	24.19
LegalBenchCorporateLobbying	66.71	75.45	72.96	72.93	34.84	74.76	69.61	75.44	76.12	73.42	**76.90**
MSMARCO-PL	4.73	7.77	8.40	**12.05**	0.00	3.72	3.29	4.72	7.23	10.34	6.98
MedicalQARetrieval	17.43	15.74	18.38	22.06	1.27	**27.00**	19.19	19.25	21.49	21.39	22.31
MedicalRetrieval	0.35	0.39	0.63	**1.35**	0.11	0.59	0.64	0.48	0.52	0.59	0.82
NFCorpus	21.90	24.74	24.90	27.01	5.50	27.67	20.49	26.97	**28.90**	28.53	28.89
NFCorpus-PL	9.88	**13.42**	9.28	6.28	4.34	9.67	8.74	8.03	13.04	12.21	10.14
NQ-PL	0.61	0.67	0.44	0.61	0.03	0.48	0.36	0.25	0.63	**0.72**	0.50
NanoNFCorpusRetrieval	18.44	30.14	28.92	26.84	5.87	**35.45**	21.15	32.39	34.33	33.74	35.36
NanoNQRetrieval	25.56	14.97	26.62	22.21	1.00	25.80	27.41	19.39	22.90	**28.20**	25.77
NanoTouche2020Retrieval	**60.09**	51.60	51.82	40.28	14.74	50.83	48.99	50.89	52.39	53.72	53.85
NeuCLIR2023RetrievalHardNegatives	1.60	**7.61**	2.02	0.19	0.23	1.03	0.50	4.44	3.01	5.14	2.20
PublicHealthQA	15.30	**20.09**	19.37	8.15	8.05	17.53	13.65	16.41	18.20	17.37	18.86
SCIDOCS	11.11	13.45	14.03	13.21	2.45	13.79	12.15	11.69	13.66	14.42	**15.93**
SciFact-PL	3.43	**12.39**	5.79	3.26	1.07	4.09	2.86	9.27	9.80	9.39	8.72
SlovakSumRetrieval	12.80	29.59	16.52	8.47	5.56	12.52	12.13	16.87	**29.86**	29.50	27.50
TRECCOVID	50.94	49.40	49.69	**59.93**	21.94	48.96	47.00	48.58	55.40	52.93	51.89
TV2Nordretrieval	11.27	**29.44**	13.84	9.41	4.42	10.21	10.26	18.58	26.48	27.48	24.42
TurHistQuadRetrieval	10.35	14.10	11.29	10.75	3.97	11.84	8.86	8.83	16.19	**16.95**	14.52
TwitterHjerneRetrieval	20.86	22.39	18.48	21.97	17.10	18.74	19.37	**23.01**	20.10	19.22	20.22
VideoRetrieval	5.96	5.72	7.10	**8.59**	3.31	7.47	6.92	5.23	6.19	6.28	7.11
VieQuADRetrieval	4.03	**9.44**	6.24	2.95	3.03	4.10	4.51	5.00	7.32	8.36	7.01
XMarket	7.77	6.96	8.90	**13.21**	2.31	10.44	6.94	5.54	8.62	8.52	9.45
45 Avg. MRR@10	16.90	18.38	17.09	16.21	7.27	17.12	15.63	17.30	19.30	**19.41**	19.27

TABLE II-7

THIS TABLE REPORTS THE PERFORMANCE OF JTCSE AND DERIVED MODELS AND OTHER BASELINES THROUGH THE MTEB PACKAGE ON 45 TEXT RETRIEVAL TASKS EVALUATED WITH THE METRIC **NDCG@1**.

Tasks	Sim∼	ESim∼	Diff∼	Info∼	SN∼	Whinen∼	Rank∼	TN∼	JT∼	JT∼ D	JT∼ UCD
AILACasedocs	**24.00**	**24.00**	12.00	16.00	14.00	10.00	14.00	16.00	16.00	16.00	18.00
AlloprofRetrieval	2.38	3.37	1.34	1.17	0.43	0.60	1.99	2.33	3.45	**3.67**	3.02
CQADupstackAndroidRetrieval	16.88	17.74	18.17	23.18	1.86	21.75	18.46	17.74	20.46	20.46	**24.61**
ClimateFEVER	9.71	5.80	8.73	6.19	2.54	9.84	9.45	8.21	7.95	8.86	**9.90**
ClimateFEVERHardNegatives	**10.00**	6.00	8.60	5.80	2.70	9.60	9.80	8.40	7.80	9.20	9.80
CodeFeedbackMT	19.30	23.42	22.87	22.56	12.77	21.91	19.99	20.25	21.77	21.35	**23.59**
CodeTransOceanDL	**8.33**	7.78	**8.33**	**8.33**	**8.33**	**8.33**	7.78	**8.33**	**8.33**	**8.33**	**8.33**
CosQA	1.60	1.80	2.20	**3.80**	0.00	1.60	1.80	1.20	2.60	2.80	2.20
CrossLingualSemanticDiscriminationWMT19	10.32	13.92	5.64	6.45	6.11	6.52	4.35	**16.63**	14.60	13.37	8.83
CrossLingualSemanticDiscriminationWMT21	15.45	21.17	12.99	10.08	9.63	11.31	11.09	**25.42**	22.06	19.49	16.35
DBPedia-PLHardNegatives	**11.00**	10.13	7.88	8.13	1.38	7.25	7.63	8.13	9.75	10.63	9.50
DanFeverRetrieval	19.32	18.41	19.60	19.63	10.70	**20.84**	19.39	14.19	18.83	18.59	19.16
DuRetrieval	1.10	2.30	1.70	2.60	0.20	2.25	1.95	2.15	3.10	2.95	**3.20**
EcomRetrieval	1.10	0.90	1.40	**2.50**	0.20	**2.50**	1.70	1.30	1.70	2.10	2.30
FEVER	15.18	5.10	12.84	8.94	2.30	**16.19**	12.41	10.22	8.66	10.02	11.21
GeorgianFAQRetrieval	1.79	1.64	1.75	2.11	0.78	2.18	1.44	1.21	2.22	**2.61**	2.30
GerDaLIR	0.55	0.63	0.46	0.39	0.25	0.42	0.40	0.41	0.67	**0.74**	0.73
GerDaLIRSmall	1.37	1.53	1.02	0.95	0.58	1.05	0.95	1.06	1.64	**1.76**	1.75
GermanQuAD-Retrieval	24.09	**29.90**	18.24	14.75	5.31	11.16	22.28	20.83	28.86	28.00	27.27
GreekCivicsQA	2.95	**3.93**	2.95	0.98	0.49	2.70	2.46	2.21	3.19	**3.93**	**3.93**
LEMBNarrativeQARetrieval	7.13	6.27	7.55	6.24	1.35	7.73	7.01	8.13	**8.14**	7.63	7.94
LEMBSummScreenFDRetrieval	32.44	36.61	**39.29**	32.44	11.31	38.99	27.98	37.80	36.31	37.80	38.39
LeCaRDv2	24.53	18.24	25.16	**27.67**	25.16	20.76	22.64	16.35	22.01	20.13	18.87
LegalBenchCorporateLobbying	55.59	64.41	62.65	62.65	25.29	65.88	59.41	65.59	**67.06**	63.53	**67.06**
MSMARCO-PL	2.33	1.94	**5.81**	3.49	0.00	2.33	2.33	1.55	2.71	5.04	3.49
MedicalQARetrieval	10.50	9.91	11.38	14.01	0.49	**17.97**	12.26	11.82	13.09	13.18	13.48
MedicalRetrieval	0.20	0.30	0.40	**1.10**	0.00	0.40	0.50	0.30	0.30	0.40	0.60
NFCorpus	14.55	17.18	15.02	17.49	2.17	17.18	12.69	17.49	**19.51**	19.04	18.89
NFCorpus-PL	5.73	8.05	5.42	2.32	2.01	5.11	5.26	4.49	**8.20**	7.12	5.42
NQ-PL	0.32	0.26	0.17	**0.41**	0.03	0.35	0.17	0.17	0.32	0.38	0.23
NanoNFCorpusRetrieval	14.00	24.00	22.00	20.00	2.00	**28.00**	16.00	20.00	26.00	24.00	**28.00**
NanoNQRetrieval	14.00	4.00	**16.00**	12.00	0.00	**16.00**	**16.00**	10.00	10.00	**16.00**	12.00
NanoTouche2020Retrieval	**42.86**	34.69	30.61	20.41	8.16	28.57	28.57	30.61	30.61	30.61	32.65
NeuCLIR2023RetrievalHardNegatives	0.00	**4.51**	1.35	0.00	0.00	0.00	0.00	1.35	0.90	0.90	0.00
PublicHealthQA	8.14	**12.79**	**12.79**	3.49	3.49	10.47	8.14	9.30	10.47	9.30	9.30
SCIDOCS	6.20	7.70	8.60	7.80	1.80	7.90	7.30	6.90	8.50	9.30	**10.20**
SciFact-PL	1.67	**9.67**	4.67	2.33	0.67	3.33	1.33	7.00	6.33	6.33	5.33
SlovakSumRetrieval	8.50	**24.33**	10.83	5.67	3.33	8.17	8.33	12.17	23.83	23.67	21.50
TRECCOVID	38.00	32.00	36.00	**44.00**	9.00	35.00	32.00	33.00	**44.00**	42.00	38.00
TV2Nordretrieval	8.30	**23.19**	10.50	6.74	3.03	7.72	7.28	13.72	20.51	21.44	18.21
TurHistQuadRetrieval	6.64	8.30	7.32	6.93	2.34	7.42	5.57	6.15	10.55	**11.13**	10.06
TwitterHjerneRetrieval	14.10	**16.67**	12.82	14.10	11.54	12.82	11.54	15.39	14.10	14.10	15.39
VideoRetrieval	4.40	4.10	5.40	**6.20**	2.20	5.50	4.90	3.50	4.40	4.30	5.20
VieQuADRetrieval	1.86	4.79	3.27	1.51	1.71	2.30	2.54	3.03	4.20	**4.83**	4.20
XMarket	4.41	4.06	5.52	**8.79**	1.49	6.94	3.67	3.57	5.47	5.18	6.02
45 Avg. NDCG@1	11.62	12.83	11.76	10.94	4.42	11.66	10.50	11.68	13.36	**13.38**	13.25

TABLE II-8

THIS TABLE REPORTS THE PERFORMANCE OF JTCSE AND DERIVED MODELS AND OTHER BASELINES THROUGH THE MTEB PACKAGE ON 45 TEXT RETRIEVAL TASKS EVALUATED WITH THE METRIC **NDCG@5**.

Tasks	Sim∼	ESim∼	Diff∼	Info∼	SN∼	Whinen∼	Rank∼	TN∼	JT∼	JT∼ D	JT∼ UCD
AILACasedocs	20.69	19.81	12.84	20.25	14.93	12.36	15.46	17.14	19.07	19.63	**21.29**
AlloprofRetrieval	4.42	**6.86**	2.55	2.63	0.90	1.37	3.73	4.22	6.34	6.31	5.69
CQADupstackAndroidRetrieval	20.54	20.72	22.20	27.26	2.51	24.82	21.03	20.47	23.20	23.72	**28.38**
ClimateFEVER	10.43	6.91	9.89	6.06	2.53	10.66	**11.32**	9.82	8.52	9.65	10.87
ClimateFEVERHardNegatives	11.26	7.64	10.31	6.56	2.50	11.07	**11.79**	10.35	9.24	10.14	11.34
CodeFeedbackMT	23.89	28.47	27.73	27.41	16.54	26.97	24.88	25.17	26.93	26.58	**29.14**
CodeTransOceanDL	18.09	19.00	19.49	16.89	20.86	**21.83**	16.23	20.64	19.95	18.58	19.60
CosQA	3.37	3.69	4.33	**7.72**	0.19	3.01	4.27	2.22	4.21	4.50	4.95
CrossLingualSemanticDiscriminationWMT19	16.95	20.78	10.07	10.44	10.24	11.27	8.01	**25.72**	22.03	20.62	14.56
CrossLingualSemanticDiscriminationWMT21	24.71	33.24	20.30	18.72	15.20	19.68	18.43	**39.74**	35.11	32.53	26.27
DBPedia-PLHardNegatives	**9.08**	8.10	7.80	7.28	1.26	7.54	7.92	5.79	8.69	9.07	8.67
DanFeverRetrieval	23.40	23.34	24.26	23.02	12.63	**24.89**	23.74	17.91	23.72	23.65	24.26
DuRetrieval	0.97	1.83	1.46	2.13	0.17	1.91	1.50	1.80	2.54	**2.63**	2.59
EcomRetrieval	2.09	1.74	2.76	**4.12**	0.69	3.66	3.06	2.49	2.82	3.22	3.77
FEVER	22.52	8.24	20.16	12.51	3.82	**24.13**	18.84	15.58	13.37	15.16	16.87
GeorgianFAQRetrieval	2.92	2.57	3.14	3.47	1.37	3.54	2.46	2.38	3.78	**4.19**	4.11
GerDaLIR	0.72	0.89	0.59	0.56	0.30	0.55	0.53	0.63	0.96	1.00	**1.02**
GerDaLIRSmall	1.92	2.09	1.38	1.31	0.84	1.40	1.41	1.61	2.28	2.30	**2.32**
GermanQuAD-Retrieval	33.11	**40.36**	26.27	21.86	9.18	17.74	32.00	30.20	39.39	38.57	37.58
GreekCivicsQA	6.39	6.45	5.82	2.32	1.19	6.59	4.89	3.71	7.30	8.07	**8.07**
LEMBNarrativeQARetrieval	10.29	9.25	10.95	9.48	2.62	11.14	10.40	**12.09**	11.87	11.27	11.50
LEMBSummScreenFDRetrieval	50.42	51.97	54.77	48.17	21.12	**56.37**	43.37	54.08	53.40	53.26	53.26
LeCaRDv2	17.60	12.58	17.39	**19.78**	18.20	13.68	16.94	10.39	16.01	12.95	12.93
LegalBenchCorporateLobbying	70.37	78.27	75.69	75.40	37.83	77.15	72.30	78.26	78.97	76.69	**79.84**
MSMARCO-PL	1.67	2.44	3.23	**4.25**	0.00	1.30	1.76	1.60	1.97	2.67	2.65
MedicalQARetrieval	18.94	16.94	19.75	23.63	1.37	**29.25**	20.78	21.07	23.16	23.09	24.33
MedicalRetrieval	0.35	0.42	0.65	**1.41**	0.13	0.65	0.59	0.43	0.52	0.66	0.81
NFCorpus	11.78	12.80	13.21	13.43	1.84	15.85	10.48	15.20	16.24	15.79	**16.67**
NFCorpus-PL	5.03	**7.54**	4.86	2.79	1.40	5.77	4.13	4.25	7.10	6.80	5.98
NQ-PL	0.60	0.63	0.41	0.60	0.03	0.41	0.35	0.20	0.59	**0.70**	0.50
NanoNFCorpusRetrieval	9.15	17.10	18.45	13.12	2.43	**21.22**	11.03	20.71	20.54	20.32	19.90
NanoNQRetrieval	26.97	18.04	**29.11**	23.03	1.26	27.28	29.09	20.07	25.89	28.56	28.84
NanoTouche2020Retrieval	33.73	28.80	32.50	22.93	5.02	34.64	30.50	30.74	35.06	**38.76**	33.28
NeuCLIR2023RetrievalHardNegatives	0.54	**2.68**	0.46	0.00	0.00	0.13	0.21	1.93	0.68	1.89	0.77
PublicHealthQA	16.85	21.12	20.80	8.59	8.83	18.61	15.54	18.36	19.90	19.98	**22.09**
SCIDOCS	4.70	5.74	5.95	5.70	0.93	6.00	4.97	4.90	5.92	6.26	**6.69**
SciFact-PL	3.53	**11.90**	5.52	3.45	1.23	4.05	2.99	9.59	9.92	9.83	8.83
SlovakSumRetrieval	13.61	30.53	17.43	8.81	5.95	13.01	12.77	17.70	**31.14**	30.67	28.56
TRECCOVID	28.78	27.57	29.54	**39.84**	8.60	24.83	29.42	26.52	30.84	30.18	31.70
TV2Nordretrieval	11.70	**30.83**	14.50	9.90	4.66	10.63	10.86	19.75	27.71	28.83	25.72
TurHistQuadRetrieval	6.98	9.50	7.49	7.17	2.66	7.88	5.73	5.83	10.88	**11.31**	9.65
TwitterHjerneRetrieval	11.40	11.49	9.86	**11.58**	9.13	9.90	10.72	11.50	10.18	9.26	10.43
VideoRetrieval	6.15	6.04	7.56	**8.90**	3.46	7.85	7.13	5.52	6.42	6.66	7.59
VieQuADRetrieval	2.65	**6.50**	4.15	1.98	1.99	2.70	3.05	3.28	4.84	5.68	4.73
XMarket	4.52	3.97	5.37	**8.13**	1.15	6.21	3.99	3.08	5.04	5.07	5.62
45 Avg. NDCG@5	13.91	15.27	14.29	13.21	5.77	14.25	13.12	14.55	16.32	**16.38**	16.32

TABLE II-9

THIS TABLE REPORTS THE PERFORMANCE OF JTCSE AND DERIVED MODELS AND OTHER BASELINES THROUGH THE MTEB PACKAGE ON 45 TEXT RETRIEVAL TASKS EVALUATED WITH THE METRIC **NDCG@10**.

Tasks	Sim∼	ESim∼	Diff∼	Info∼	SN∼	Whinen∼	Rank∼	TN∼	JT∼	JT∼ D	JT∼ UCD
AILACasedocs	23.09	**23.92**	16.63	21.80	17.33	13.78	19.71	18.72	19.95	20.45	22.41
AlloprofRetrieval	5.29	**8.02**	3.23	3.44	1.16	1.82	4.73	5.17	7.58	7.66	6.93
CQADupstackAndroidRetrieval	22.33	22.38	23.77	29.56	2.82	27.48	22.67	22.47	25.20	25.61	**30.06**
ClimateFEVER	12.60	8.58	11.93	7.36	3.12	12.87	**13.40**	12.14	10.91	12.05	13.32
ClimateFEVERHardNegatives	13.54	9.66	12.59	8.21	3.21	13.62	**14.13**	12.95	11.74	12.95	13.75
CodeFeedbackMT	25.15	29.78	28.93	28.67	17.61	28.07	26.02	26.40	28.23	27.89	**30.51**
CodeTransOceanDL	29.61	30.30	31.54	26.69	32.57	33.35	25.63	**33.94**	30.66	28.33	29.83
CosQA	4.27	4.83	5.36	**9.42**	0.31	3.85	4.92	2.76	5.25	5.88	6.12
CrossLingualSemanticDiscriminationWMT19	18.51	22.43	11.64	11.85	11.24	12.48	9.38	**27.89**	23.73	22.07	16.13
CrossLingualSemanticDiscriminationWMT21	26.73	35.64	22.18	20.15	18.01	21.80	20.90	**42.40**	36.85	34.93	28.63
DBPedia-PLHardNegatives	**9.00**	8.32	7.80	7.07	1.09	7.42	7.95	5.74	8.60	8.63	8.40
DanFeverRetrieval	24.31	24.40	25.13	23.72	13.11	**25.78**	24.51	18.65	24.81	24.73	25.35
DuRetrieval	1.15	2.06	1.53	2.33	0.18	2.05	1.61	1.97	2.67	**2.81**	2.81
EcomRetrieval	2.51	2.18	3.23	**4.81**	0.72	4.08	3.45	3.01	3.22	3.57	4.06
FEVER	24.82	9.82	22.44	13.78	4.44	**26.67**	20.88	17.52	15.02	17.01	19.03
GeorgianFAQRetrieval	3.32	2.96	3.60	3.98	1.62	4.03	2.95	2.74	4.41	**4.74**	4.69
GerDaLIR	0.83	1.00	0.64	0.61	0.35	0.62	0.64	0.72	1.07	1.13	**1.13**
GerDaLIRSmall	2.23	2.37	1.54	1.45	0.96	1.57	1.65	1.83	2.56	2.56	**2.62**
GermanQuAD-Retrieval	35.61	**42.57**	28.25	24.25	10.48	20.21	33.98	32.55	41.93	40.74	40.08
GreekCivicsQA	8.55	8.12	7.61	3.05	1.68	8.08	6.37	4.91	8.90	**10.21**	10.15
LEMBNarrativeQARetrieval	11.52	10.47	12.16	10.70	3.44	12.27	11.75	**13.71**	13.30	12.74	12.90
LEMBSummScreenFDRetrieval	54.17	55.37	58.27	52.82	24.39	**60.37**	47.15	58.43	57.85	57.08	56.83
LeCaRDv2	15.02	10.28	14.46	**16.94**	15.10	11.53	13.77	8.61	13.15	10.11	10.58
LegalBenchCorporateLobbying	71.70	80.10	77.68	77.90	39.76	78.96	74.48	79.66	80.37	77.90	**81.26**
MSMARCO-PL	1.98	2.51	3.03	**3.97**	0.00	1.22	1.54	1.33	1.70	2.52	1.93
MedicalQARetrieval	21.66	19.29	22.68	26.80	1.82	**31.90**	23.13	23.60	26.39	26.32	27.40
MedicalRetrieval	0.48	0.49	0.81	**1.50**	0.16	0.71	0.79	0.62	0.72	0.75	0.98
NFCorpus	10.47	12.16	12.16	12.32	1.92	14.41	9.96	13.49	**15.24**	14.65	14.91
NFCorpus-PL	4.69	6.88	4.94	2.77	1.46	5.40	3.77	3.84	**6.96**	6.23	5.67
NQ-PL	0.70	0.80	0.53	0.68	0.03	0.51	0.39	0.28	0.74	**0.83**	0.62
NanoNFCorpusRetrieval	8.27	15.01	15.88	12.45	3.14	**18.18**	9.86	17.62	17.85	17.27	16.98
NanoNQRetrieval	33.44	20.57	32.29	27.60	1.26	31.53	34.31	24.62	28.99	**34.44**	33.58
NanoTouche2020Retrieval	32.39	29.28	30.66	21.90	5.05	34.50	30.88	29.62	33.20	**35.38**	34.88
NeuCLIR2023RetrievalHardNegatives	0.97	**2.92**	0.59	0.03	0.11	0.42	0.19	1.69	1.49	2.44	0.93
PublicHealthQA	20.13	**24.37**	23.84	10.89	11.06	22.40	17.05	20.52	22.20	21.83	23.59
SCIDOCS	5.70	7.15	7.41	6.99	1.09	7.21	5.99	6.33	7.45	7.57	**8.23**
SciFact-PL	4.51	**13.67**	6.24	3.76	1.23	4.29	3.82	9.82	10.90	10.49	10.29
SlovakSumRetrieval	15.69	32.65	19.97	10.41	6.96	15.42	14.56	19.98	**33.32**	32.76	31.15
TRECCOVID	24.82	26.07	26.19	**35.13**	6.56	22.26	27.37	23.54	27.10	27.35	28.83
TV2Nordretrieval	13.04	**33.04**	15.87	10.98	5.34	11.72	12.07	21.36	30.02	30.97	28.10
TurHistQuadRetrieval	8.08	11.14	8.86	8.19	3.19	9.12	6.96	6.51	12.45	**12.96**	11.03
TwitterHjerneRetrieval	12.55	12.71	11.93	12.81	10.21	11.84	12.03	**14.46**	12.06	10.80	11.85
VideoRetrieval	6.96	6.69	8.18	**10.11**	3.88	8.66	8.16	6.23	7.30	7.46	8.27
VieQuADRetrieval	3.54	**8.11**	5.32	2.42	2.45	3.28	3.65	3.91	6.16	6.98	5.70
XMarket	4.80	4.13	5.41	**8.13**	1.14	6.30	4.19	3.11	5.15	5.17	5.71
45 Avg. NDCG@10	15.13	16.56	15.44	14.23	6.51	15.42	14.30	15.72	17.45	17.49	**17.52**

TABLE II-10

THIS TABLE REPORTS THE PERFORMANCE OF JTCSE AND DERIVED MODELS AND OTHER BASELINES THROUGH THE MTEB PACKAGE ON 45 TEXT RETRIEVAL TASKS EVALUATED WITH THE METRIC **PRECISION@1**.

Tasks	Sim∼	ESim∼	Diff∼	Info∼	SN∼	Whinen∼	Rank∼	TN∼	JT∼	JT∼ D	JT∼ UCD
AILACasedocs	**24.00**	**24.00**	12.00	16.00	14.00	10.00	14.00	16.00	16.00	16.00	18.00
AlloprofRetrieval	2.38	3.37	1.34	1.17	0.43	0.60	1.99	2.33	3.45	**3.67**	3.02
CQADupstackAndroidRetrieval	16.88	17.74	18.17	23.18	1.86	21.75	18.46	17.74	20.46	20.46	**24.61**
ClimateFEVER	9.71	5.80	8.73	6.19	2.54	9.84	9.45	8.21	7.95	8.86	**9.90**
ClimateFEVERHardNegatives	**10.00**	6.00	8.60	5.80	2.70	9.60	9.80	8.40	7.80	9.20	9.80
CodeFeedbackMT	19.30	23.42	22.87	22.56	12.77	21.91	19.99	20.25	21.77	21.35	**23.59**
CodeTransOceanDL	**8.33**	7.78	**8.33**	**8.33**	**8.33**	**8.33**	7.78	**8.33**	**8.33**	**8.33**	**8.33**
CosQA	1.60	1.80	2.20	**3.80**	0.00	1.60	1.80	1.20	2.60	2.80	2.20
CrossLingualSemanticDiscriminationWMT19	10.32	13.92	5.64	6.45	6.11	6.52	4.35	**16.63**	14.60	13.37	8.83
CrossLingualSemanticDiscriminationWMT21	15.45	21.17	12.99	10.08	9.63	11.31	11.09	**25.42**	22.06	19.49	16.35
DBPedia-PLHardNegatives	14.75	13.75	10.75	11.50	2.25	9.50	10.25	12.00	14.25	**15.50**	13.50
DanFeverRetrieval	19.32	18.41	19.60	19.63	10.70	**20.84**	19.39	14.19	18.83	18.59	19.16
DuRetrieval	1.10	2.30	1.70	2.60	0.20	2.25	1.95	2.15	3.10	2.95	**3.20**
EcomRetrieval	1.10	0.90	1.40	**2.50**	0.20	**2.50**	1.70	1.30	1.70	2.10	2.30
FEVER	15.18	5.10	12.84	8.94	2.30	**16.19**	12.41	10.22	8.66	10.02	11.21
GeorgianFAQRetrieval	1.79	1.64	1.75	2.11	0.78	2.18	1.44	1.21	2.22	**2.61**	2.30
GerDaLIR	0.55	0.63	0.46	0.39	0.25	0.42	0.40	0.41	0.67	**0.74**	0.73
GerDaLIRSmall	1.37	1.53	1.02	0.95	0.58	1.05	0.95	1.06	1.64	**1.76**	1.75
GermanQuAD-Retrieval	24.09	**29.90**	18.24	14.75	5.31	11.16	22.28	20.83	28.86	28.00	27.27
GreekCivicsQA	2.95	**3.93**	2.95	0.98	0.49	2.70	2.46	2.21	3.19	**3.93**	**3.93**
LEMBNarrativeQARetrieval	7.13	6.27	7.55	6.24	1.35	7.73	7.01	8.13	**8.14**	7.63	7.94
LEMBSummScreenFDRetrieval	32.44	36.61	**39.29**	32.44	11.31	38.99	27.98	37.80	36.31	37.80	38.39
LeCaRDv2	24.53	18.24	25.16	**27.67**	25.16	20.76	22.64	16.35	22.01	20.13	18.87
LegalBenchCorporateLobbying	55.59	64.41	62.65	62.65	25.29	65.88	59.41	65.59	**67.06**	63.53	**67.06**
MSMARCO-PL	2.33	4.65	**6.98**	**6.98**	0.00	2.33	2.33	2.33	4.65	**6.98**	4.65
MedicalQARetrieval	10.50	9.91	11.38	14.01	0.49	**17.97**	12.26	11.82	13.09	13.18	13.48
MedicalRetrieval	0.20	0.30	0.40	**1.10**	0.00	0.40	0.50	0.30	0.30	0.40	0.60
NFCorpus	15.48	18.27	17.03	18.89	2.79	18.27	13.93	19.20	**20.74**	20.43	**20.74**
NFCorpus-PL	6.19	**8.67**	5.88	2.79	2.48	5.26	5.57	4.95	**8.67**	7.74	5.88
NQ-PL	0.32	0.26	0.17	**0.41**	0.03	0.35	0.17	0.17	0.32	0.38	0.23
NanoNFCorpusRetrieval	14.00	24.00	22.00	20.00	2.00	**28.00**	16.00	20.00	26.00	24.00	**28.00**
NanoNQRetrieval	14.00	4.00	**16.00**	12.00	0.00	**16.00**	**16.00**	10.00	10.00	**16.00**	12.00
NanoTouche2020Retrieval	**42.86**	34.69	30.61	20.41	8.16	28.57	28.57	30.61	30.61	30.61	32.65
NeuCLIR2023RetrievalHardNegatives	0.00	**5.41**	1.35	0.00	0.00	0.00	0.00	2.70	1.35	1.35	0.00
PublicHealthQA	8.14	**12.79**	**12.79**	3.49	3.49	10.47	8.14	9.30	10.47	9.30	9.30
SCIDOCS	6.20	7.70	8.60	7.80	1.80	7.90	7.30	6.90	8.50	9.30	**10.20**
SciFact-PL	1.67	**9.67**	4.67	2.33	0.67	3.33	1.33	7.00	6.33	6.33	5.33
SlovakSumRetrieval	8.50	**24.33**	10.83	5.67	3.33	8.17	8.33	12.17	23.83	23.67	21.50
TRECCOVID	42.00	36.00	40.00	**46.00**	10.00	40.00	34.00	36.00	**46.00**	44.00	40.00
TV2Nordretrieval	8.30	**23.19**	10.50	6.74	3.03	7.72	7.28	13.72	20.51	21.44	18.21
TurHistQuadRetrieval	6.64	8.30	7.32	6.93	2.34	7.42	5.57	6.15	10.55	**11.13**	10.06
TwitterHjerneRetrieval	14.10	**16.67**	12.82	14.10	11.54	12.82	11.54	15.39	14.10	14.10	15.39
VideoRetrieval	4.40	4.10	5.40	**6.20**	2.20	5.50	4.90	3.50	4.40	4.30	5.20
VieQuADRetrieval	1.86	4.79	3.27	1.51	1.71	2.30	2.54	3.03	4.20	**4.83**	4.20
XMarket	4.41	4.06	5.52	**8.79**	1.49	6.94	3.67	3.57	5.47	5.18	6.02
45 Avg. PRECISION@1	11.82	13.12	11.99	11.18	4.49	11.85	10.64	11.93	13.59	**13.63**	13.46

TABLE II-11

THIS TABLE REPORTS THE PERFORMANCE OF JTCSE AND DERIVED MODELS AND OTHER BASELINES THROUGH THE MTEB PACKAGE ON 45 TEXT RETRIEVAL TASKS EVALUATED WITH THE METRIC **PRECISION@5**.

Tasks	Sim∼	ESim∼	Diff∼	Info∼	SN∼	Whinen∼	Rank∼	TN∼	JT∼	JT∼ D	JT∼ UCD
AILACasedocs	13.20	12.40	9.60	14.00	10.40	9.20	10.80	11.60	12.40	12.80	**14.80**
AlloprofRetrieval	1.29	**2.02**	0.74	0.81	0.29	0.42	1.09	1.23	1.84	1.79	1.65
CQADupstackAndroidRetrieval	7.15	6.75	7.41	9.01	1.03	7.87	6.81	6.95	7.70	7.90	**9.44**
ClimateFEVER	5.71	3.99	5.51	3.26	1.42	5.85	**6.48**	5.80	4.93	5.52	6.07
ClimateFEVERHardNegatives	6.20	4.44	5.72	3.58	1.36	6.14	**6.68**	6.12	5.40	5.82	6.44
CodeFeedbackMT	5.58	6.58	6.40	6.33	3.98	6.27	5.83	5.89	6.30	6.23	**6.79**
CodeTransOceanDL	6.00	6.56	6.44	5.44	7.00	**7.44**	5.33	7.11	6.78	6.22	6.67
CosQA	1.04	1.20	1.32	**2.40**	0.08	0.88	1.44	0.72	1.20	1.28	1.56
CrossLingualSemanticDiscriminationWMT19	4.62	5.47	2.89	2.84	2.82	3.19	2.32	**6.86**	5.84	5.47	3.99
CrossLingualSemanticDiscriminationWMT21	6.70	8.89	5.47	5.38	4.14	5.55	5.15	**10.71**	9.56	8.91	7.10
DBPedia-PLHardNegatives	10.00	9.70	9.05	9.20	1.75	8.85	9.30	7.00	**10.85**	10.80	10.55
DanFeverRetrieval	5.38	5.50	5.67	5.19	2.87	5.67	5.49	4.23	5.56	5.58	**5.70**
DuRetrieval	0.71	1.21	0.97	1.43	0.14	1.23	0.96	1.11	1.65	**1.76**	1.69
EcomRetrieval	0.60	0.50	0.82	**1.12**	0.22	0.94	0.84	0.72	0.78	0.88	1.02
FEVER	6.32	2.41	5.80	3.39	1.13	**6.80**	5.33	4.49	3.86	4.34	4.77
GeorgianFAQRetrieval	0.81	0.69	0.87	0.94	0.37	0.98	0.69	0.69	1.05	**1.15**	1.14
GerDaLIR	0.20	0.26	0.16	0.17	0.09	0.15	0.15	0.19	0.28	0.28	**0.29**
GerDaLIRSmall	0.56	0.59	0.38	0.38	0.25	0.40	0.42	0.49	**0.65**	0.64	**0.65**
GermanQuAD-Retrieval	8.27	**9.88**	6.66	5.69	2.52	4.80	8.12	7.76	9.72	9.56	9.33
GreekCivicsQA	1.92	1.87	1.77	0.74	0.34	2.02	1.43	1.03	2.21	2.36	**2.41**
LEMBNarrativeQARetrieval	2.65	2.41	2.82	2.51	0.78	2.86	2.70	**3.17**	3.08	2.92	2.96
LEMBSummScreenFDRetrieval	13.21	13.21	13.63	12.32	6.13	**14.23**	11.37	13.75	13.63	13.51	13.33
LeCaRDv2	15.98	11.20	15.60	**17.74**	16.35	11.95	15.35	8.93	14.59	11.57	11.57
LegalBenchCorporateLobbying	16.59	17.77	17.24	17.18	9.82	17.29	16.59	17.65	17.82	17.59	**18.06**
MSMARCO-PL	1.86	4.19	3.26	**6.98**	0.00	1.40	1.86	1.86	2.33	2.79	2.79
MedicalQARetrieval	5.37	4.73	5.54	6.46	0.44	**7.86**	5.69	5.96	6.49	6.49	6.88
MedicalRetrieval	0.10	0.12	0.18	**0.34**	0.04	0.18	0.14	0.10	0.16	0.20	0.20
NFCorpus	11.08	11.33	12.26	12.07	1.92	14.30	9.91	13.87	14.86	14.49	**15.73**
NFCorpus-PL	4.64	**6.50**	4.40	2.72	1.49	5.20	3.72	4.15	6.13	5.82	5.51
NQ-PL	0.21	**0.24**	0.15	0.17	0.01	0.12	0.13	0.06	0.19	0.23	0.17
NanoNFCorpusRetrieval	8.40	14.40	15.60	10.80	2.40	17.60	9.60	**18.00**	17.20	17.60	16.40
NanoNQRetrieval	8.00	6.80	8.40	7.20	0.40	7.60	8.40	5.60	8.00	8.00	**9.20**
NanoTouche2020Retrieval	31.84	27.76	31.84	23.27	4.08	35.10	30.61	30.61	36.33	**40.41**	33.06
NeuCLIR2023RetrievalHardNegatives	1.08	**3.24**	0.27	0.00	0.00	0.27	0.27	2.97	1.08	2.43	1.08
PublicHealthQA	5.12	5.58	5.81	2.56	2.79	5.35	4.65	5.35	5.58	6.05	**6.74**
SCIDOCS	4.20	5.12	5.22	5.08	0.76	5.26	4.28	4.36	5.18	5.42	**5.64**
SciFact-PL	1.07	**3.07**	1.47	0.93	0.40	1.07	1.07	2.73	2.93	2.93	2.53
SlovakSumRetrieval	3.73	7.20	4.67	2.37	1.67	3.50	3.40	4.60	**7.53**	7.37	7.00
TRECCOVID	29.20	30.80	31.60	**40.80**	9.60	26.40	31.60	27.20	30.80	30.40	33.60
TV2Nordretrieval	2.93	**7.57**	3.63	2.54	1.25	2.65	2.83	5.07	6.87	7.12	6.51
TurHistQuadRetrieval	3.20	4.45	3.40	3.22	1.27	3.56	2.56	2.52	4.90	**5.06**	4.28
TwitterHjerneRetrieval	7.44	7.69	6.92	**8.46**	6.67	6.67	7.44	7.44	6.67	5.90	7.18
VideoRetrieval	1.54	1.54	1.94	**2.28**	0.92	2.02	1.80	1.48	1.64	1.76	1.98
VieQuADRetrieval	1.34	**3.28**	2.05	0.98	0.95	1.29	1.47	1.49	2.30	2.73	2.24
XMarket	3.40	3.01	3.92	**5.88**	0.91	4.42	3.10	2.24	3.69	3.75	4.13
45 Avg. PRECISION@5	6.14	6.54	6.34	6.14	2.52	6.28	5.89	6.26	7.08	**7.15**	7.13

TABLE II-12

THIS TABLE REPORTS THE PERFORMANCE OF JTCSE AND DERIVED MODELS AND OTHER BASELINES THROUGH THE MTEB PACKAGE ON 45 TEXT RETRIEVAL TASKS EVALUATED WITH THE METRIC **PRECISION@10**.

Tasks	Sim∼	ESim∼	Diff∼	Info∼	SN∼	Whinen∼	Rank∼	TN∼	JT∼	JT∼ D	JT∼ UCD
AILACasedocs	**10.20**	**10.20**	8.20	9.40	8.00	6.80	9.80	8.80	8.80	8.80	9.80
AlloprofRetrieval	0.92	**1.38**	0.58	0.66	0.23	0.35	0.86	0.91	1.30	1.32	1.21
CQADupstackAndroidRetrieval	4.56	4.32	4.55	**5.75**	0.67	5.24	4.39	4.49	4.89	4.99	5.74
ClimateFEVER	4.28	3.04	4.07	2.46	1.06	4.42	4.63	4.42	3.96	4.32	**4.65**
ClimateFEVERHardNegatives	4.64	3.47	4.29	2.82	1.14	4.74	**4.86**	4.74	4.30	4.69	**4.86**
CodeFeedbackMT	3.18	3.70	3.57	3.55	2.32	3.47	3.26	3.33	3.55	3.52	**3.82**
CodeTransOceanDL	6.61	6.72	6.94	5.78	7.17	7.28	5.61	**7.67**	6.67	6.11	6.50
CosQA	0.80	0.96	0.98	**1.72**	0.08	0.70	0.92	0.52	0.92	1.06	1.14
CrossLingualSemanticDiscriminationWMT19	2.80	3.25	1.93	1.86	1.72	1.97	1.60	**4.10**	3.44	3.18	2.48
CrossLingualSemanticDiscriminationWMT21	3.95	5.19	3.32	3.12	2.96	3.44	3.34	**6.18**	5.32	5.20	4.28
DBPedia-PLHardNegatives	8.40	8.58	7.93	7.60	1.30	7.30	7.83	5.90	**9.00**	8.58	8.60
DanFeverRetrieval	2.97	3.08	3.10	2.81	1.58	3.11	2.98	2.35	3.12	3.12	**3.19**
DuRetrieval	0.56	0.90	0.63	1.03	0.10	0.83	0.66	0.80	1.13	**1.24**	1.21
EcomRetrieval	0.43	0.39	0.56	**0.77**	0.12	0.60	0.54	0.52	0.51	0.55	0.60
FEVER	3.92	1.73	3.68	2.11	0.78	**4.26**	3.34	2.90	2.49	2.78	3.10
GeorgianFAQRetrieval	0.53	0.47	0.57	0.63	0.27	0.64	0.50	0.46	0.72	**0.75**	**0.75**
GerDaLIR	0.14	0.16	0.09	0.10	0.06	0.10	0.11	0.13	0.18	**0.19**	0.18
GerDaLIRSmall	0.38	0.39	0.25	0.24	0.17	0.26	0.29	0.32	0.42	0.40	**0.43**
GermanQuAD-Retrieval	4.91	5.63	3.95	3.59	1.67	3.17	4.66	4.61	**5.65**	5.45	5.45
GreekCivicsQA	1.62	1.45	1.45	0.59	0.32	1.47	1.18	0.89	1.60	**1.87**	1.84
LEMBNarrativeQARetrieval	1.71	1.59	1.79	1.63	0.65	1.79	1.77	**2.09**	1.98	1.92	1.92
LEMBSummScreenFDRetrieval	7.77	7.68	7.89	7.59	4.08	**8.33**	6.88	8.21	8.19	7.92	7.77
LeCaRDv2	13.08	8.55	12.20	**14.53**	12.77	9.75	11.45	7.11	11.20	8.18	8.87
LegalBenchCorporateLobbying	8.71	9.44	9.24	9.35	5.53	9.21	8.97	9.27	9.35	9.18	**9.47**
MSMARCO-PL	2.79	3.72	3.26	**5.58**	0.00	1.40	1.63	1.40	2.09	3.02	1.86
MedicalQARetrieval	3.52	3.09	3.67	4.20	0.36	**4.75**	3.57	3.76	4.24	4.23	4.38
MedicalRetrieval	0.09	0.08	0.14	**0.20**	0.03	0.11	0.13	0.11	0.14	0.13	0.15
NFCorpus	8.45	9.85	9.60	9.54	1.80	11.12	8.20	10.56	**12.32**	11.67	11.70
NFCorpus-PL	3.81	5.11	3.87	2.45	1.46	4.27	2.97	3.19	**5.33**	4.61	4.61
NQ-PL	0.14	**0.18**	0.12	0.12	0.00	0.10	0.08	0.06	0.16	0.17	0.13
NanoNFCorpusRetrieval	7.60	11.60	12.00	10.40	2.60	13.40	7.60	13.00	**13.80**	13.20	12.60
NanoNQRetrieval	**6.20**	4.20	5.20	5.00	0.20	5.20	5.80	4.40	5.00	5.80	6.00
NanoTouche2020Retrieval	29.18	27.55	27.96	21.02	4.49	32.86	28.98	27.35	31.22	32.86	**33.67**
NeuCLIR2023RetrievalHardNegatives	1.49	**3.11**	0.81	0.14	0.14	0.95	0.27	2.16	2.30	2.97	1.08
PublicHealthQA	3.61	**3.84**	**3.84**	1.98	2.09	**3.84**	2.79	3.37	3.49	3.61	**3.84**
SCIDOCS	3.01	3.84	3.92	3.71	0.53	3.74	3.07	3.48	3.98	3.90	**4.22**
SciFact-PL	0.83	**2.20**	0.97	0.57	0.20	0.60	0.80	1.43	1.77	1.67	1.77
SlovakSumRetrieval	2.52	4.25	3.12	1.68	1.15	2.50	2.25	3.02	**4.45**	4.32	4.30
TRECCOVID	24.80	28.40	26.80	**35.20**	6.80	23.20	29.00	24.60	27.40	28.20	30.00
TV2Nordretrieval	1.88	**4.46**	2.25	1.61	0.84	1.66	1.79	3.03	4.15	4.22	4.00
TurHistQuadRetrieval	2.16	3.07	2.38	2.13	0.90	2.40	1.90	1.61	3.24	**3.36**	2.83
TwitterHjerneRetrieval	4.62	4.87	5.00	5.39	4.23	4.87	4.87	**5.90**	4.62	3.97	4.74
VideoRetrieval	1.02	0.97	1.16	**1.52**	0.59	1.26	1.23	0.96	1.09	1.13	1.20
VieQuADRetrieval	1.13	**2.45**	1.63	0.71	0.71	0.94	1.04	1.06	1.82	2.03	1.61
XMarket	2.87	2.52	3.03	**4.48**	0.76	3.45	2.62	1.81	2.94	2.99	3.25
45 Avg. PRECISION@10	4.64	4.92	4.72	4.61	1.88	4.71	4.47	4.60	5.20	5.19	**5.24**

TABLE II-13

THIS TABLE REPORTS THE PERFORMANCE OF JTCSE AND DERIVED MODELS AND OTHER BASELINES THROUGH THE MTEB PACKAGE ON 45 TEXT RETRIEVAL TASKS EVALUATED WITH THE METRIC **RECALL@1**.

Tasks	Sim∼	ESim∼	Diff∼	Info∼	SN∼	Whinen∼	Rank∼	TN∼	JT∼	JT∼ D	JT∼ UCD
AILACasedocs	7.91	**9.23**	5.53	4.49	4.51	3.37	5.46	5.96	7.70	6.03	6.16
AlloprofRetrieval	2.38	3.37	1.34	1.17	0.43	0.60	1.99	2.33	3.45	**3.67**	3.02
CQADupstackAndroidRetrieval	12.78	13.98	14.83	18.32	1.27	17.16	14.30	13.33	15.87	15.99	**19.14**
ClimateFEVER	4.58	2.50	4.22	2.71	1.10	4.57	**4.61**	3.54	3.43	3.85	4.32
ClimateFEVERHardNegatives	**4.93**	2.54	4.32	2.80	1.13	4.57	4.85	3.67	3.46	4.04	4.34
CodeFeedbackMT	19.30	23.42	22.87	22.56	12.77	21.91	19.99	20.25	21.77	21.35	**23.59**
CodeTransOceanDL	**8.33**	7.78	**8.33**	**8.33**	**8.33**	**8.33**	7.78	**8.33**	**8.33**	**8.33**	**8.33**
CosQA	1.60	1.80	2.20	**3.80**	0.00	1.60	1.80	1.20	2.60	2.80	2.20
CrossLingualSemanticDiscriminationWMT19	10.32	13.92	5.64	6.45	6.11	6.52	4.35	**16.63**	14.60	13.37	8.83
CrossLingualSemanticDiscriminationWMT21	15.45	21.17	12.99	10.08	9.63	11.31	11.09	**25.42**	22.06	19.49	16.35
DBPedia-PLHardNegatives	**1.96**	1.08	0.80	0.70	0.05	1.20	0.93	0.79	1.11	1.11	0.80
DanFeverRetrieval	19.32	18.41	19.59	19.63	10.70	**20.84**	19.39	14.19	18.83	18.59	19.16
DuRetrieval	0.22	0.70	0.46	0.73	0.03	0.67	0.57	0.71	**1.03**	0.96	1.00
EcomRetrieval	1.10	0.90	1.40	**2.50**	0.20	**2.50**	1.70	1.30	1.70	2.10	2.30
FEVER	14.35	4.80	12.13	8.43	2.21	**15.30**	11.73	9.48	8.03	9.26	10.43
GeorgianFAQRetrieval	1.79	1.64	1.75	2.11	0.78	2.18	1.44	1.21	2.22	**2.61**	2.30
GerDaLIR	0.49	0.56	0.43	0.34	0.22	0.39	0.36	0.37	0.61	0.67	**0.67**
GerDaLIRSmall	1.22	1.38	0.92	0.85	0.51	0.96	0.87	0.95	1.51	**1.61**	1.58
GermanQuAD-Retrieval	24.09	**29.90**	18.24	14.75	5.31	11.16	22.28	20.83	28.86	28.00	27.27
GreekCivicsQA	2.95	**3.93**	2.95	0.98	0.49	2.70	2.46	2.21	3.19	**3.93**	**3.93**
LEMBNarrativeQARetrieval	7.13	6.27	7.55	6.24	1.35	7.73	7.01	8.13	**8.14**	7.63	7.94
LEMBSummScreenFDRetrieval	32.44	36.61	**39.29**	32.44	11.31	38.99	27.98	37.80	36.31	37.80	38.39
LeCaRDv2	0.92	0.72	1.01	**1.20**	0.92	0.77	0.84	0.59	0.84	0.78	0.69
LegalBenchCorporateLobbying	55.59	64.41	62.65	62.65	25.29	65.88	59.41	65.59	**67.06**	63.53	**67.06**
MSMARCO-PL	0.02	0.04	0.05	0.16	0.00	0.02	0.01	0.10	0.13	**0.21**	0.11
MedicalQARetrieval	10.50	9.91	11.38	14.01	0.49	**17.97**	12.26	11.82	13.09	13.18	13.48
MedicalRetrieval	0.20	0.30	0.40	**1.10**	0.00	0.40	0.50	0.30	0.30	0.40	0.60
NFCorpus	0.73	1.18	1.27	1.26	0.06	1.26	0.91	1.59	**1.76**	1.44	1.36
NFCorpus-PL	0.38	1.03	0.67	0.23	0.10	0.48	0.63	0.48	**1.12**	0.80	0.48
NQ-PL	0.24	0.20	0.15	**0.38**	0.03	0.27	0.14	0.14	0.26	0.33	0.20
NanoNFCorpusRetrieval	0.38	1.64	2.59	1.59	0.00	**3.74**	0.49	3.59	1.63	1.60	1.88
NanoNQRetrieval	14.00	4.00	**16.00**	11.00	0.00	**16.00**	**16.00**	10.00	10.00	15.00	12.00
NanoTouche2020Retrieval	**2.97**	2.55	2.33	1.31	0.49	2.20	2.45	2.29	2.38	2.33	2.27
NeuCLIR2023RetrievalHardNegatives	0.00	**0.30**	0.01	0.00	0.00	0.00	0.00	0.02	0.01	0.01	0.00
PublicHealthQA	8.14	**12.79**	**12.79**	3.49	3.49	10.47	8.14	9.30	10.47	9.30	9.30
SCIDOCS	1.26	1.58	1.76	1.60	0.37	1.62	1.49	1.42	1.73	1.89	**2.09**
SciFact-PL	1.67	**9.03**	4.28	2.33	0.67	3.17	1.33	6.67	5.53	5.94	4.78
SlovakSumRetrieval	8.50	**24.33**	10.83	5.67	3.33	8.17	8.33	12.17	23.83	23.67	21.50
TRECCOVID	0.10	0.10	0.10	**0.13**	0.03	0.11	0.09	0.08	0.12	0.11	0.11
TV2Nordretrieval	8.30	**23.19**	10.50	6.74	3.03	7.72	7.28	13.72	20.51	21.44	18.21
TurHistQuadRetrieval	3.32	4.15	3.66	3.47	1.17	3.71	2.78	3.08	5.27	**5.57**	5.03
TwitterHjerneRetrieval	3.93	4.57	3.33	3.61	3.36	4.00	3.14	**4.74**	4.42	4.32	4.36
VideoRetrieval	4.40	4.10	5.40	**6.20**	2.20	5.50	4.90	3.50	4.40	4.30	5.20
VieQuADRetrieval	0.93	2.39	1.64	0.76	0.85	1.15	1.27	1.51	2.10	**2.42**	2.10
XMarket	1.06	0.80	1.32	**2.14**	0.17	1.64	0.64	0.70	1.21	1.21	1.41
45 Avg. RECALL@1	7.16	8.43	7.60	6.70	2.77	7.57	6.80	7.82	**8.73**	**8.73**	8.58

TABLE II-14

THIS TABLE REPORTS THE PERFORMANCE OF JTCSE AND DERIVED MODELS AND OTHER BASELINES THROUGH THE MTEB PACKAGE ON 45 TEXT RETRIEVAL TASKS EVALUATED WITH THE METRIC **RECALL@5**.

Tasks	Sim∼	ESim∼	Diff∼	Info∼	SN∼	Whinen∼	Rank∼	TN∼	JT∼	JT∼ D	JT∼ UCD
AILACasedocs	18.77	17.20	11.68	**21.96**	14.49	11.25	14.16	17.44	18.19	20.60	21.48
AlloprofRetrieval	6.43	**10.10**	3.71	4.06	1.43	2.07	5.44	6.13	9.20	8.94	8.25
CQADupstackAndroidRetrieval	23.86	23.87	26.19	31.41	3.03	27.98	24.46	23.89	27.00	27.97	**32.66**
ClimateFEVER	11.69	8.20	11.43	6.82	2.81	12.18	**13.18**	11.71	9.89	11.15	12.42
ClimateFEVERHardNegatives	12.85	9.18	11.97	7.65	2.62	12.71	**13.81**	12.49	11.16	11.89	13.52
CodeFeedbackMT	27.92	32.91	31.98	31.64	19.88	31.33	29.13	29.46	31.50	31.15	**33.93**
CodeTransOceanDL	30.00	32.78	32.22	27.22	35.00	**37.22**	26.67	35.56	33.89	31.11	33.33
CosQA	5.20	6.00	6.60	**12.00**	0.40	4.40	7.20	3.60	6.00	6.40	7.80
CrossLingualSemanticDiscriminationWMT19	23.08	27.36	14.46	14.19	14.12	15.95	11.61	**34.28**	29.19	27.36	19.96
CrossLingualSemanticDiscriminationWMT21	33.48	44.46	27.32	26.88	20.72	27.77	25.76	**53.53**	47.82	44.57	35.50
DBPedia-PLHardNegatives	3.77	2.76	3.80	2.84	0.37	3.07	**3.90**	2.01	3.21	3.61	3.70
DanFeverRetrieval	26.90	27.51	28.32	25.94	14.34	28.34	27.44	21.17	27.81	27.88	**28.51**
DuRetrieval	0.99	1.61	1.39	1.92	0.17	1.81	1.31	1.66	2.29	**2.39**	2.34
EcomRetrieval	3.00	2.50	4.10	**5.60**	1.10	4.70	4.20	3.60	3.90	4.40	5.10
FEVER	29.35	11.32	27.05	15.88	5.26	**31.65**	24.96	20.81	17.86	20.07	22.12
GeorgianFAQRetrieval	4.06	3.47	4.33	4.72	1.87	4.87	3.43	3.43	5.26	**5.77**	5.69
GerDaLIR	0.91	1.15	0.71	0.74	0.37	0.68	0.67	0.85	1.26	1.27	**1.31**
GerDaLIRSmall	2.45	2.67	1.75	1.68	1.09	1.78	1.85	2.16	**2.87**	2.82	2.85
GermanQuAD-Retrieval	41.33	**49.41**	33.30	28.45	12.61	24.00	40.61	38.79	48.59	47.78	46.64
GreekCivicsQA	9.58	9.34	8.85	3.69	1.72	10.07	7.13	5.16	11.06	11.79	**12.04**
LEMBNarrativeQARetrieval	13.26	12.07	14.09	12.55	3.90	14.31	13.50	**15.83**	15.41	14.61	14.82
LEMBSummScreenFDRetrieval	66.07	66.07	68.16	61.61	30.66	**71.13**	56.85	68.75	68.16	67.56	66.67
LeCaRDv2	3.16	2.09	3.00	**3.34**	3.10	2.18	3.02	1.59	2.64	2.14	2.09
LegalBenchCorporateLobbying	82.94	88.82	86.18	85.88	49.12	86.47	82.94	88.24	89.12	87.94	**90.29**
MSMARCO-PL	0.05	0.18	0.20	**0.48**	0.00	0.07	0.05	0.17	0.25	0.27	0.24
MedicalQARetrieval	26.86	23.63	27.69	32.28	2.20	**39.31**	28.47	29.79	32.47	32.47	34.38
MedicalRetrieval	0.50	0.60	0.90	**1.70**	0.20	0.90	0.70	0.50	0.80	1.00	1.00
NFCorpus	2.95	2.76	3.71	3.99	0.40	**5.00**	2.72	4.72	4.35	4.08	4.68
NFCorpus-PL	1.42	2.18	1.65	0.88	0.28	2.47	1.20	0.99	**2.60**	2.59	2.59
NQ-PL	0.89	0.98	0.62	0.76	0.03	0.50	0.51	0.25	0.85	**1.01**	0.77
NanoNFCorpusRetrieval	1.11	4.57	5.04	2.75	0.26	5.58	2.08	5.74	5.57	5.83	**5.98**
NanoNQRetrieval	39.00	32.00	41.00	34.00	2.00	38.00	41.00	28.00	39.00	39.00	**45.00**
NanoTouche2020Retrieval	10.94	9.36	10.64	6.88	1.10	12.31	11.05	10.14	12.08	**13.69**	10.81
NeuCLIR2023RetrievalHardNegatives	0.04	0.37	0.01	0.00	0.00	0.01	0.07	**0.57**	0.05	0.17	0.16
PublicHealthQA	25.58	27.91	29.07	12.79	13.95	26.74	23.26	26.74	27.91	30.23	**33.72**
SCIDOCS	4.26	5.20	5.29	5.17	0.77	5.35	4.38	4.43	5.25	5.50	**5.72**
SciFact-PL	5.17	**14.14**	6.39	4.50	1.83	4.78	4.78	11.68	13.04	13.04	11.72
SlovakSumRetrieval	18.67	36.00	23.33	11.83	8.33	17.50	17.00	23.00	**37.67**	36.83	35.00
TRECCOVID	0.35	0.35	0.38	**0.51**	0.12	0.32	0.39	0.37	0.37	0.39	0.42
TV2Nordretrieval	14.65	**37.84**	18.16	12.70	6.25	13.23	14.16	25.34	34.33	35.60	32.57
TurHistQuadRetrieval	8.01	11.13	8.50	8.06	3.17	8.89	6.40	6.30	12.26	**12.65**	10.69
TwitterHjerneRetrieval	11.07	10.24	9.55	**11.13**	8.87	9.47	10.56	10.11	9.04	7.80	9.59
VideoRetrieval	7.70	7.70	9.70	**11.40**	4.60	10.10	9.00	7.40	8.20	8.80	9.90
VieQuADRetrieval	3.35	**8.20**	5.13	2.44	2.37	3.22	3.66	3.74	5.74	6.84	5.59
XMarket	2.98	2.50	3.63	**5.44**	0.53	4.12	2.65	1.85	3.25	3.33	3.58
45 Avg. RECALL@5	14.81	16.24	14.96	13.52	6.61	15.02	13.94	15.64	17.30	17.38	**17.49**

TABLE II-15
THIS TABLE REPORTS THE PERFORMANCE OF JTCSE AND DERIVED MODELS AND OTHER BASELINES THROUGH THE MTEB PACKAGE ON 45 TEXT RETRIEVAL TASKS EVALUATED WITH THE METRIC **RECALL@10**.

Tasks	Sim∼	ESim∼	Diff∼	Info∼	SN∼	Whinen∼	Rank∼	TN∼	JT∼	JT∼ D	JT∼ UCD
AILACasedocs	28.03	**31.16**	22.53	28.94	22.83	16.82	26.91	23.56	22.89	25.47	28.48
AlloprofRetrieval	9.15	**13.77**	5.83	6.61	2.25	3.50	8.59	9.11	13.04	13.17	12.05
CQADupstackAndroidRetrieval	29.43	29.00	31.21	**38.47**	4.15	36.20	29.43	30.25	33.10	33.88	37.96
ClimateFEVER	16.56	12.09	16.02	9.81	4.22	17.19	17.78	16.92	15.40	16.61	**17.95**
ClimateFEVERHardNegatives	18.03	13.92	17.17	11.56	4.25	18.57	**19.12**	18.42	16.88	18.35	18.85
CodeFeedbackMT	31.81	36.97	35.69	35.54	23.21	34.74	32.63	33.27	35.53	35.21	**38.18**
CodeTransOceanDL	66.11	67.22	69.44	57.78	71.67	72.78	56.11	**76.67**	66.67	61.11	65.00
CosQA	8.00	9.60	9.80	**17.20**	0.80	7.00	9.20	5.20	9.20	10.60	11.40
CrossLingualSemanticDiscriminationWMT19	27.97	32.45	19.28	18.60	17.24	19.69	15.95	**41.01**	34.42	31.84	24.78
CrossLingualSemanticDiscriminationWMT21	39.53	51.85	33.15	31.24	29.56	34.38	33.37	**61.81**	53.19	51.96	42.78
DBPedia-PLHardNegatives	**5.63**	5.18	5.38	4.01	0.51	4.37	5.53	3.45	5.15	5.21	5.24
DanFeverRetrieval	29.72	30.77	31.01	28.09	15.83	31.06	29.84	23.46	31.19	31.23	**31.90**
DuRetrieval	1.53	2.35	1.74	2.64	0.23	2.36	1.82	2.24	2.86	**3.10**	3.09
EcomRetrieval	4.30	3.90	5.60	**7.70**	1.20	6.00	5.40	5.20	5.10	5.50	6.00
FEVER	36.30	16.17	33.93	19.74	7.18	**39.30**	31.15	26.65	22.89	25.64	28.71
GeorgianFAQRetrieval	5.34	4.68	5.73	6.28	2.65	6.39	4.95	4.56	7.21	**7.49**	**7.49**
GerDaLIR	1.25	1.48	0.86	0.89	0.50	0.88	1.00	1.13	1.59	**1.67**	1.65
GerDaLIRSmall	3.37	3.52	2.24	2.10	1.48	2.29	2.59	2.83	3.73	3.60	**3.77**
GermanQuAD-Retrieval	49.05	56.26	39.47	35.89	16.70	31.67	46.64	46.05	**56.53**	54.49	54.45
GreekCivicsQA	16.22	14.50	14.50	5.90	3.19	14.74	11.79	8.85	15.97	**18.67**	18.43
LEMBNarrativeQARetrieval	17.07	15.86	17.86	16.32	6.48	17.86	17.70	**20.85**	19.84	19.15	19.15
LEMBSummScreenFDRetrieval	77.68	76.79	78.87	75.89	40.77	**83.33**	68.75	82.14	81.85	79.17	77.68
LeCaRDv2	5.05	3.23	4.55	**5.39**	4.80	3.62	4.58	2.58	4.10	2.98	3.26
LegalBenchCorporateLobbying	87.06	94.41	92.35	93.53	55.29	92.06	89.71	92.65	93.53	91.77	**94.71**
MSMARCO-PL	0.37	0.35	0.40	**0.80**	0.00	0.16	0.17	0.26	0.38	0.49	0.36
MedicalQARetrieval	35.21	30.86	36.72	42.04	3.61	**47.46**	35.74	37.55	42.43	42.33	43.80
MedicalRetrieval	0.90	0.80	1.40	**2.00**	0.30	1.10	1.30	1.10	1.40	1.30	1.50
NFCorpus	4.60	4.80	5.56	5.83	0.73	6.82	4.61	5.86	6.58	6.56	**7.03**
NFCorpus-PL	2.10	3.17	2.88	1.27	0.40	3.00	1.53	1.94	**3.36**	3.26	3.23
NQ-PL	1.20	**1.49**	0.98	1.01	0.03	0.82	0.64	0.48	1.30	1.39	1.13
NanoNFCorpusRetrieval	1.76	5.70	5.66	3.92	2.78	6.83	4.46	**8.00**	6.90	6.65	6.84
NanoNQRetrieval	**59.00**	40.00	51.00	48.00	2.00	50.00	57.00	42.00	49.00	57.00	**59.00**
NanoTouche2020Retrieval	19.05	17.49	18.35	12.29	2.54	21.29	19.53	17.20	20.14	21.29	**21.48**
NeuCLIR2023RetrievalHardNegatives	0.26	**1.22**	0.07	0.01	0.01	0.08	0.08	0.62	1.05	1.05	0.64
PublicHealthQA	36.05	**38.37**	**38.37**	19.77	20.93	**38.37**	27.91	33.72	34.88	36.05	**38.37**
SCIDOCS	6.09	7.81	7.95	7.53	1.07	7.58	6.26	7.05	8.07	7.91	**8.56**
SciFact-PL	8.17	**19.46**	8.72	5.50	1.83	5.44	7.28	12.34	16.04	15.04	16.12
SlovakSumRetrieval	25.17	42.50	31.17	16.83	11.50	25.00	22.50	30.17	**44.50**	43.17	43.00
TRECCOVID	0.58	0.63	0.63	**0.85**	0.16	0.56	0.69	0.63	0.64	0.67	0.71
TV2Nordretrieval	18.80	**44.63**	22.46	16.06	8.35	16.60	17.92	30.27	41.46	42.19	39.99
TurHistQuadRetrieval	10.79	15.33	11.91	10.65	4.49	12.01	9.47	8.06	16.21	**16.80**	14.16
TwitterHjerneRetrieval	13.40	13.10	14.02	14.17	11.35	13.65	13.27	**16.94**	13.72	11.69	12.78
VideoRetrieval	10.20	9.70	11.60	**15.20**	5.90	12.60	12.30	9.60	10.90	11.30	12.00
VieQuADRetrieval	5.64	**12.26**	8.13	3.56	3.54	4.69	5.20	5.32	9.11	10.16	8.06
XMarket	4.50	3.72	4.99	**7.50**	0.80	5.85	3.99	2.63	4.63	4.72	5.16
45 Avg. RECALL@10	19.51	20.90	19.49	17.66	9.32	19.48	18.28	20.24	21.88	21.97	**22.15**